I. Drug Dosage

The drugs in the ER and WARD of pediatrics

注意事項: 一定要問過敏史與 G6PDdef, Penicillin 類要做 PCT (<3m/o 不用做)

若劑量不確定,可以 40kg 計為成人劑量,視比例調整,如 20kg 則藥物為成人半量,且勿超過成人最大劑量. 但仍建議先查詢藥典,如:micromedex, UpToDate,院內藥典等,尤其特殊用藥更須注意劑量

Newborn 的劑量不同,要另外查

注意國小前或小於 7 歲前兒童可能不會吞藥,須開立糖漿

注意小於 20kg 或 8 歲兒童,每種藥物通常不超過 10ml/dose. 腹瀉時勿用太多糖漿,易拉肚子

注意所有糖漿加起來 c.c 數是否難以餵藥

1.Fever : Antipyretics are not recommended under 3 m/o(小於 3 個月嬰兒不要規則吃退燒藥,應視病況而定)

Fever routine:
>38.0℃→ ice packing (< 3m/0 → water pillow)
>38.5℃ → acetaminophen or ibuprofen
>39 ℃ →voltaren
>39.5℃→ warm water bath

Acetaminophen: (最安全的第一線用藥)	1~2 ml/kg/day q6h *(=BW/2 ml q6h)* 10-15 mg/kg/dose q6h	24mg/ml 60ml/pc
Ibuprofen (< 6m/o 禁用)	1~2 ml/kg/day q6h *(=BW/2 ml q6h)* 5-10 mg/kg/dose q6h	400mg/tab 20mg/ml, 60ml/bot
Voltaren supp:	0.1-0.8 mg/kg/dose q6h prn 15~20kg→ 1# , 10~14kg→ 2/3#	12.5 mg/supp(一天不超過 3 次) 7.5~9kg→ 1/2# , <7.5kg→ 1/3#
Indomethacin supp:	1.5-2.5mg/kg/day, Max 4mg/kg/day 20kg: 1/2#, 30kg:3/4#, >40kg:1# q6h prn	50mg/supp (腹瀉及 hema 病人不建議用塞劑)

2.Cough:

Secorine:	1~2 ml/kg/day qid *(=BW/4ml qid)*	120ml/pc
Robitussin (Guaifenesin):	1~2ml/kg/day qid *(=BW/4ml qid)*	120ml/pc
Medicon-A :	Adult 1# tid	
Medicon syrup : (<2yr,少用)	10kg---1ml po q6h *(=BW/8 ml/kg/dose qid)*	1ml= dextromethorphan 2.5mg, potassium cresolsulfonate 15mg
Ambroxol :	40kg— 1# tid < 1 ml/kg/day <2yrs: 7.5mg bid*(2.5ml bid)* 2-6yrs: 7.5mg tid.*(2.5ml tid)* 6-12yrs: 15mg bid-tid *(5ml bid-tid)*	30mg/tab 3mg/ml, 100ml/bot

Cough mixture:(CMA)	1~2 ml/kg/day qid *(=BW/4 ml qid)*	120ml/bot, 60ml/bot 1ml= platycodon fluid extract 0.2ml (platycodon extract 65mg), sugar contents 30%.

3.Rhinorrhea/Nasal obstruction:

Periactin (cyproheptadine): 鼻水	< 1 ml/kg/day qid *(=BW/4ml qid)*	4mg/tab; 0.4mg/ml, 60ml/bot(<3m/o 儘不開,易昏睡)
Spiz : (peace):鼻塞 (<1y/o 儘不開,易興奮)	< 1 ml/kg/day qid *(=BW/4ml qid)*	60ml/bot, 1ml=6mg pseudoephedrine+0.25mg trprolidine
Clarinase :	>12 yr : 1# Bid	1# =5mg loratadine + 120mg pseudoephedrine
Levocetirizine: (Xyzal)	6mo~5yr: 1.25mg HS 6~11 yr: 2.5mg HS >12 yr: 5mg HS	5mg/tab
Cetirizine oral soln (Zytec)	2-6 y/o : 2.5mg HS-Q12H >6y/o: 5mg HS-Q12H	1mg/ml

4.Urticaria:

Periactin (cyproheptadine):	< 1 ml/kg/day qid *(=BW/4ml qid)*	4mg/tab; 0.4mg/ml, 60ml/bot
Diphenhydramine (Vena) (drug-induced EPS 解藥)	1mg/kg/dose IV or IM q6h prn < 6yr: 6.25~12.5mg q4-6h PO (Max 150mg/day)	30mg/ml/amp 50mg/cap

5.GI:

MgO	Adult 1~2# qid 20kg 0.5# qid	250mg/tab
AMG	1-2ml/kg/day q2h--qid *(=BW/2 ml qid)*	80ml/bot; 200ml/bot
Bio three (Clostridium Butyricum 10mg + Streptococcus faecalis 2mg)	3~6Kg ---1/3 # po tid 6~10Kg---1/2# po tid 10~15Kg --- 1# po tid	15-25Kg--- 1.5# tid >25Kg --- 2# tid
Lactobacillus rhamnosus (摩舒益多膠囊)	1# qd	150mg/cap
Buscopan (Hyoscine) (小兒科少用)	IV,IM: 0.5mg/kg/dose PO: 0.5mg/kg/dose qid	20mg/ml/amp 10mg/tab.
Cimetidine (Tagamet)	PO: 5-10mg/kg/dose q6h IM,IVF: 5~10mg/kg/dose q6~8h	200mg/tab 200mg/2ml/amp (IVF over 15-20min)

Famotidine (Gaster)	PO: 0.5mg/kg/dose qd~q12h IVF: 0.25mg/kg/dose q12h	20mg/tab 20mg/amp
Esomeprazole	IVF, PO: 1mg/kg/dose qd~q12h Adult 20mg qd~q12h	40mg/vial IV (in 100ml N/S run > 30mins) 40mg/tab PO
Gascon (Dimethicone)	Child : individualized (1~3ml tid) *(=BW/10 tid)* Adult: 2-4ml tid	20mg/ml, 30ml/bot
Dulcolax supp.	Supp: 0-10yr 1/2# supp >10yr 1# supp	10mg/supp PO: 5mg/tab (AC)
Fleet enema	5ml/kg/dose Max :1pc	118ml/bot
Kaopectin :	1~2 ml/kg/day qid*(=BW/4 qid)*	120ml/pc(須與其他藥隔 15-30 分吃)
Smecta	PO: < 1yr 1pc/day tid 1~2 yr 2pc/day tid >2yr 3pc/day tid (同 adult)	3gm/pc in 50cc water 與其他藥隔 1.5hr for acute & chronic diarrhea
Imodium (Loperamide)小兒科少用	PO: 1mg/10kg	2mg/cap
Novamin :	PO: 0.1mg/kg/dose po qid IM: 0.2 mg/kg/dose IM	5mg/tab 5mg/ml/amp(<2yr,少用)
Primperan (Metoclopramide) (AC) (注意 EPS)	<5yr 盡量不用 IM,IV: 0.1mg/kg/dose, max< 0.1mg/kg/tid PO: 0.1-0.2 ml/kg/dose tid	7.68mg/2ml/amp IV 3.84mg/tab PO 1mg/ml, 60ml/bot PO (side effect: EPS.→Vena 1mg/kg/dose IV or IM

6.Bronchial asthma:

O2 & suction prn		
Inhalation drug ↓		
Bricanyl (Terbutaline)	0.2 mg/kg q20mins x3 次(max<10mg)	5mg/2ml/vial
Atrovent (Ipratropium)	0.25~0.5mg q20mins x3 次	0.5mg/2ml/vial
Bosmin (epinephrine)		1mg/ml/amp
Pulmicort (Budesonide)		1mg/2ml/respule
Ventolin(Salbutamol)	0.15mg/kg then 0.05~0.15mg/kg q20~30mins (max<5mg)	 100mcg/ds
Subcutaneous drug↓		
Bosmin (epinephrine) (S.C.)	0.01ml/kg/dose **S.C** (1:1000) (max<0.3ml/dose) q20mins x3 次	1mg/ml/amp (1:1000)
Oral drug ↓		
Berotec (Fenoterol)	PO: < 1ml/kg/day qid *(=BW/4 qid)*	2.5mg/5ml, 60ml/bot, 2.5mg/tab
Meptin (Procaterol)	PO: < 0.5ml/kg/day Bid	5 ug/ml, 60ml/bot, 25 ug/tab

Salbutamol	**PO**: 2~6 yr 0.1~0.2mg/kg/dose tid 6~12 yr 2mg/dose tid >12yr 2~4mgdose tid	2mg/tab
IV steroid ↓		
Hydrocortisone sodium (Solu-cortef)	IV(快),IM(慢): 1-5mg/kg/day qd-q12h	100mg/vial
Dexan (Dexamethasone)	0.25-0.5mg/kg/day IV q6h	5mg/1ml/amp; 0.5mg/tab ; 4mg/tab
Methylprednisolone (Solu-medrol)	1~2mg/kg/day IV q6h	40mg/vial
Oral steroid ↓		
Prednisolone	PO: 1-2mg/kg/day, qid (Max: <60mg/day)	5mg/tab 1mg/ml, 60 ml/bot

7.Convulsion:

O2 supply: keep airway patent		
↓		
If new case, check: CBC/DC , B/C , BUN , Cr ,GPT, GOT , Na , K , Ca , P , Mg, ABG, Sugar , Drug level , ect .		
↓		
Set IV line		
Valium(Diazepam)	Sooner onset (10mg/2ml/amp) <0.3 mg/kg/dose IV slowly push *(=BW/4 mg st)* 0.5 mg/kg/dose SUPP *(=BW/2 mg st)* If convulsion doesn't stop in 5 min, Give another dose. <5yr, max<5mg; >5yr, max<10mg, repeat q2h-q4h	
↓		
Phenbarbital (Luminal) 易影響 Conscious 注意 Allergy, apnea	Later onset Loading dose :10~**20**mg/kg/dose IV slow push/IVF (max<300mg) Maintain dose: 5mg/kg/day IVF q12h Bolus dose 5mg/kg/dose IVF	30mg/tab; 100mg/ml/amp
Phenytoin(Dilantin) (不可與葡萄糖液混和,會結晶) 注意 Allergy	Loading dose 10~20mg/kg/dose IV 250mg/5ml/amp(in N/S) slow push/IVF 100mg/tab Maintain dose 5mg/kg/day IVF q12h Bolus dose 5mg/kg/dose IVF	250mg/5ml/amp(in N/S) 100mg/tab
Depakine(Valproate) <2yr 少用, 肝毒性 注意 AST/ALT/Ammonia	Loading/Bolus dose 15mg/kg/dose ivf Maintain dose 10mg/kg/dose q8h ivf	200mg/tab,500mg/tab. 200mg/ml,40ml/bot 400mg/vial

8.Antibiotics (補充)

藥物濃度: Gentamicin Amikacin Vancomycin:	follow BUN/Cr, drug level every 7 days follow BUN/Cr, drug level after 4 doses, and every 7 days	Trough level:
Acyclovir	IVF:10mg/kg/dose q8h in D5W drip 1 hr PO: 20mg/kg/dose qid x 5days	250mg/vial 200mg/cap, 40mg/ml, 25ml/bot
Oseltamivir(<5 y/o) 克流感	(=3mg/kg/dose) BID < 3m/o: 12mg bid 3-5 m/o: 20mg BID (=1/4pc bid) 6-11 m/o: 25mg BID (=1/3pc bid) BW<15Kg: 30mg BID(=0.4pc bid) BW 15-24 Kg: 45mg BID(=0.6pc bid) BW 25-40Kg: 60mg BID(=0.8pc bid) BW>40Kg: 75mg BID(=1pc bid)	75mg/cap (先 Stat 領回 10pc, 再改成五天份的自備藥) If 公費, 下載"公費流感抗病毒藥物申請單". IE→部門網頁→藥學部→表單下載
Relenza(>5y/o) 瑞樂沙	2 puff(10mg) bid x 5 days	

9. Sedative (都須填寫同意書)

(IE→醫療事務類→麻醉科→)

Chloral hydrate	25mg/kg/dose (max<100mg/kg/day or 2g/day) *(=BW/2 ml PO stat, max: 1ml/kg/day or 2g/day, 20ml/day)*	10%, 30ml/bot, 1ml=100mg **→兒童 chloral hydrate 鎮靜同意書**
Midazolam	0.1mg/kg/dose (=BW/10 mg iv push st)	5mg/1ml/amp, 15mg/3ml/amp **→中重度鎮靜同意書**
Ketamine	1mg/kg/dose (=同 BW mg iv push st)	50mg/ml, 10ml/vial **→中重度鎮靜同意書**

10.Others:

BSA= $\sqrt{[BW(kg)\times BH(cm)]/3600}$	補充水份：3000cc/ m^2 /day	
Glycerol	Continuous IVF run (BW/2) ml/hr	500ml/bot
Mannitol (↓IICP)	2.5-5ml(0.25-2g)/kg/dose IVF q6h (Max: 75ml/q6h-q4h)	20%/500ml/bot
Lasix (Furosemide)	PO: 1-4mg/kg/day QD~q6h IV: 1-2mg/kg/dose, QD~Q6H	40mg/tab, 10mg/ml, 120ml/bot 20mg/2ml/amp

Albumin	0.5~1 gm/kg/dose(=5mlxBW) IV drip 2-4 hr QD #lasix 1mg/kg/dose after albumin	20% 50ml Albumin<2.5 can regular give for 3 days 一般 Hypoalbuminemia:健保 Max 1pc/D Nephrotic syndrome: Max 2pc/Day

11.External Use

EYE:			OSMD (Sulfamethoxazole) 1-2gt qd-qid
(抗生素)			(過敏)
Acyclovir oph. oint 1cm q4h			Emadine 1gt qid (antihistamine) >3yr
Chloramphenicol(CM)(antibiotics) 1 gt q1-4h			Fluorometholone(FML) 1~2 gt bid~qid (steroid)
Tetracycline oph oint 1cm q2-4h			(人工淚液)
Tobramycin oph oint 1.25cm q8h			Duratears oph oint 1cm qd-bid
oph sol 1 gt q1~4h			Tears natural free eye drops 1-2 drops prn
EAR:	Ofloxacin Ear Drops(Tarivid)	2-3gt bid	5ml/bot (for otitis externa) 靜置 10sec 後再流出 耳膜破裂也可以點
	Sofradex Ear Drops(Anti-Corticosteroid)	2-3 gt tid~qid	8ml/bot (for otitis externa)
Oral-topical:			
Mycostatin oral susp.(Nystatin)			Kenalog oral base(Triamcinolone)口內膏 QD~Bid
100,000U/ml,24ml/bot;500,000U/tab 直接抹在口內			Benzydamine oral spray 自費(腸病毒常用)
premature 100,000U qid (1ml qid)			1 puff/ 4kg q1.5~3hr, max<4puff/dose
older infant 200,000U qid (2ml qid)			Dexamethasone oral oint 0.1%, 5g/tube
> 5 yr 500,000U qid			
Skin:			(抗生素)
Betamethasone: Rinderon(Susp. 7mg/ml/amp)			N.P.B oint bid
Fluticasone cream 每日數次 (口水疹,臉部紅疹)			Gentamicin cream bid
Tropoderm (Neo-cortisone) 每日數次 (濕疹)			Sulfasil cream bid (燙傷 有水泡的 lesion)
Elomet (mometasone) QD 使用(強力 steroid 勿抹太多)			Fucidin cream (Antibacteria)
calamine lotion tid~qid (過敏 止癢)			
Sinbaby lotion tid~qid(過敏 止癢)			
Anal:			Tropoderm cream(Neo-cortisone) 15gm/tube
Proctosedyl oint (posterisan oint) 15gm/tube			ZnO cream
Fungal:	Canesten cream (Clotrimazole)	Bid	5gm/tube
Menthol oil 抹肚子 for abdominal pain			

***** 兒科常開立檢查 ***** 請都於後面註明適應症

名稱	代碼	電話/注意事項
腦部超音波 (檢查→小兒科)	Neofetal neurosonography M32-061 (for ant. Fontanelle 未關之病人) Color-coded transcranial sonography M29-088-Z (for ant. Fontanelle 已關之病人)	8525
腦電圖 (檢查→小兒科)	EEG(sleep) < 6y/o with chloral hydrate (BW/2 ml) EEG(awake+sleep)>6y/o with chloral hydrate	8526/8525/須電話告知技術員安排時間
Brain MRI (檢查→放射診斷科→MRI)	MRI with and without contrast 需 chloral hydrate 備用 (=1ml/kg/dose)	8400/須電話告知, 常指定 W2 晚上黃敏政醫師判讀
電腦斷層 (檢查→放射診斷科→CT)	With contrast 需 Creatinine data, 及 NPO 4 hrs	復健 CT 3789 急診 CT 2568 兒童 CT 8404 /急做須電話告知
心臟超音波 (檢查→小兒科)	Color flow mapping with 2D-echo Doppler 考慮 chloral hydrate 備用(BWx0.6 ml)	8529
心電圖 (檢查→小兒科)	E.K.G (Electrocardiography) (M32-071)	
腹部超音波 (檢查→小兒科)	Abdomen echo for multiple organs or tumor (M32-063)	8240
腸套疊復位 (檢查→放射診斷科→G-I examination)	Fluoroscopic reduction of intussusception	8399
上/下消化道攝影 (檢查→放射診斷科 →G-I examination)	Upper GI series Lower GI series	8399
Gallium scan (檢查→核醫科)	Whole body inflammation scan (N73-061B)	
腎臟超音波 (檢查→小兒科)	Ped nephrosonography (M32-064)	8528
尿路迴流檢查 (檢查→放射診斷科 →G-U examination)	VCUG	8399
DMSA (檢查→核醫科)	Kidney (N73-041) + SPECT (N73-006)	
血液抹片 (檢查→小兒科)	Blood smear interpretation (M32-504)	

Bone scan (檢查→核醫科)	Bone scan (N73-060)	
胸部超音波	Chest echo gran (M21-051-Z)	8241
幫忙 On IV	PICU – BW > 10Kg, NICU – BW <10Kg, 且無傳染病的幼童	8500, 8501 3L1- 8503, 8504 3L2-8508,8509 5L1-8511, 8512 5L2-8515, 8516

兒童感染科常用抗生素

Beta-Lactams	Penicillin	Aqueous penicillin G		(1) Pneumonia:20 萬～40 萬 U/kg/day, Q4~6H (2) Meningitis: 30 萬～40 萬 U/kg/day, Q4~6H(Max: 2400 萬)
		Penicillin V		PO: 15~60mg/kg/day
		Ampicillin/ Amoxicillin		(1) Regular dose: 100~200mg/kg/day, Q6H (2) Meningitis dose: 200~400mg/kg/day, Q6H (3) Amoxicillin (PO): 20~40mg/kg/day, QID Q8H 80~90mg/kg/day for Pneumococcus
		Augmentin		(1) IV:以 Amoxicillin 計算 100mg/kg/day, Q6H (2) PO: Standard:以 Amoxicillin 計算 40~50mg/kg/day, TID High dose:以 Amoxicillin 計算 80~90mg/kg/day, BID
		Unasyn		100~300mg/kg/day, Q6H
	Penicillinase-R Penicillin	Oxacillin		150~200mg/kg/day, Q4~6H (Max: 12g/day)
		Dicloxacillin		PO: 25~50mg/kg/day, QID (Max: 500mg/dose)
	Antipseudomonal Penicillin	Piperacillin		200~300mg/kg/day, Q6H (Max:12g/day)
		Tazocin		以 Piperacillin 計算 320~480mg/kg/day(Max: 4g), Q4~6H
	Cephalosporin	1st	Cefazolin	100mg/kg/day, Q6~8H (Max: 6g/day)
			Cephalexin	PO: 20~50mg/kg/day, QID (Max: 4g/day)
			Cefadraxil	PO: 30mg/kg/day, TID (Max: 2g/day)
		2nd	Cefuroxime	(1) IV: 100mg/kg/day, Q6~8H (Max: 4.5g/day) (2) PO: 20~50mg/kg/day, BID (Max: 1g/day)
			Cefacor(Keflor)	PO: 20~40mg/kg/day, Q8H
		3rd	Cefotaxime	(1) Regular dose: 50~180mg/kg/day, Q6H (2) Meningitis dose: 200mg/kg/day, Q6H (Max: 8g/day)
			Ceftriaxone	(1) Regular dose: 50~100mg/kg/day, Q12H (2) Meningitis dose: 100mg/kg/day, Q12H (Max: 2g/day)
			Ceftazidime	(1) Regular dose: 30~50mg/kg/dose, Q6~8H (2) Meningitis dose: 200mg/kg/day (Max: 6g/day)
			Cefixime	PO: 8mg/kg/day (Max: 400mg), QD~BID
			Flomoxef	60~80mg/kg/day, Q6~8H (Max: 2g/dose Q6H)
		4th	Cefipime	Regular: 50mg/kg/dose (Max: 2g), Q12H High dose: 50mg/kg/dose, Q6H (Ps. Aeruginosa, Meningitis, Neutropenic fever)
	Monobactam	Aztreonam		30mg/kg/dose, Q8H
	Carbapenum	Imipenum+cilastatin (Tienum)		以 Imipenum 計算 15~25mg/kg/dose, Q6H(Max:2~4g/day)
		Meropenum		(1) Non-CNS: 20mg/kg/dose(Max:1g), Q8H (2) CNS: 40mg/kg/dose(Max:1g), Q8H
		Ertapenum		(1) 3m~12y: 15mg/kg/dose(Max:500mg), Q12H (2) >13y: 1g, QD

Aminoglycoside	Gentamicin	5mg/kg/day, QD (Max: 160mg)
	Amikacin	15mg/kg/day, Q8H (Max: 1.5g)
Marcolides	Erythromycin	(1) IV: 50mg/kg/day, Q6H (Max:4g/day) (2) PO: 30~50mg/kg/day, QID (Max: 2g/day)
	Clarithromycin	PO: 15mg/kg/day, BID (Max: 1g/day)
	Azithromycin	PO: 10mg/kg/day 3~5day (Max: 500mg/day)
Fluoroquinolone	Ciprofloxacin	(1) IV: 6~10mg/kg/dose, Q8H (2) PO: 10~20mg/kg/dose, BID
Glycopeptide	Vancomycin	10~15mg/kg/dose, Q6~8H
	Teicoplanin (不過 BBB)	Loading: 10mg/kg/dose, Q12H x 3dose Maintenance: 10mg/kg/dose, QD
Alternative for MRSA	Fusidic acid	PO: 10mg/kg/dose (Max: 500mg), Q12H
	Linezolid	(1) <12y: 10mg/kg/dose (Max: 600mg), Q8H (2) >12y: 10mg/kg/dose (Max: 600mg), Q12H
Tetracyclin	Doxycyclin	2.2~4.4mg/kg/day, QD (Max: 200mg/day) *小於 8 歲勿用
Antifungal	Caspofungin	70mg/m^2/dose QD x1day then 50mg/m^2/dose, QD
	Fluconazole	IV PO: 5~12mg/kg/day, QD
	Voriconazole	Loading: 6mg/kg/dose q12h x1day then 4mg/kg/dose, Q12h
	Amphotericin B	1mg/kg/day for neonatal candidiasis
	Baktar Trimethoprim-Sulfamethoxazole 以 Trimethoprim 計算	(1) UTI AOM: 8~12mg/kg/day, Q12H (2) PCJ: 20mg/kg/day, Q6H (3) UTI prophylaxis: 5mg/kg/day, QD
	Clindamycin	IV PO: 30~40mg/kg/day, Q6H
	Metronidazole	Loading: 15mg/kg/dose then 7.5mg/kg/dose, Q6h Max:4g/day
	Colistin	(1) IV: 2.5mg/kg/dose, Q12H (Max: 750mg/day) (2) INHA: 75mg (in 3cc NS), Q12H
Anti-TB	Isoniazid	10~15mg/kg/dose, QD (Max: 300mg/day)
	Rifampin	(1) For TB: 10~20mg/kg/day, QD (Max: 600mg) (2) Prophylaxis for meningococcus: 10mg/kg/dose, BIDx2d (3) Prophylaxis for H. influenzae: 20mg/kg/day, QDx4d (4) For MRSA: 10~20mg/kg/day, QD
	Pyrazinamide	15~30mg/kg/dose, QD (Max:2g/day)
	Ethanbutol	15~20mg/kg/dose, QD (Max:1.6g/day) *小於 4 歲勿用
Antiviral	Acyclovir	(1) IV: For encephalitis 20mg/kg/dose, Q8H(3m/o~12y/o) 10mg/kg/dose, Q8H(>12y/o) (2) PO: 40~80mg/kg/day, Q4H

製表：兒童感染科 陳志榮 郭貞孍 林明儒 祁孝鈞

II. Fluid

1. 量：前10kg部分： 100cc/kg/Day 約4cc/kg/hr
10~20kg部分： 50cc/kg/Day 約2cc/kg/hr
> 20kg部分： 20cc/kg/Day 約1cc/kg/hr

※ 快速記法 ：

B.W.	**10 kg =**	**1000cc/天**	**約等於**	**40 cc/hr** (20cc/hr = 1天500cc)
	20 kg =	1500cc/天		60 cc/hr
	30 kg =	1700cc/天		70 cc/hr

2. 質：① **NPO**的Maintain fluid or Normo-natremia dehydration 時,
IVF 用 D5,1/4S solution (5% Dextrose + 1/4的 N/S; 500cc/bot)或Taita Sol'n 1號 or 2號加1-2 Amp 50% G/W

② **Hypernatremia** dehydration (Na>150 meq/L, FF-up Na q6h)
至少用***half saline*** + 5%,Dextrose，每小時降Na不可超過 0.5meq/L/hr

③ **Hyponatremia** dehydration (Na< 130 meq/L, FF-up Na q12h)
每小時升Na不可超過2meq/L/hr, **Seizure** attack - 3% saline 10ml/kg/dose over 1 hr

④ **Vital signs** unstable or severe oligouria:(沒小便時勿給含K+的溶液)
Fluid challenge with N/S or Lactate-Ringer

⑤ **Hyponatremia** 脫水時，
1. Ped.Manual上Infancy脫水程度為5、10、15%, Child為3、6、9%脫水，
2. 算10%脫水時,補充Fluid**應先扣除Fluid challenge已給的fluid**,
否則前8小時等於在Fluid challenge 易造成hyponatremia , hyperglycemia (若加上2amp.50%G/W), 導致**Seizure**.

⑥ 血糖低者或持續拉肚子者，每一瓶 D51/4 S 可加 1~2 Amp 的 50%G/W(可上升glucose conc.2~4%)或參考以下台大輸液

⑦ 沒有小便時，勿加K+ 在點滴內，15%KCL sol'n勿IV push。

⑧ 小便量: 1-4cc/kg/hr (平均 2cc/kg/hr, 10kg baby 一天小便約500cc)

3. 每天體液和電解質的維持量

表一、每天體液和電解質的維持量

	體重		
	0-10公斤	10-20公斤	>20公斤
無感流失	45ml/kg	450ml+22ml/kg (超過10公斤部分)	670ml+9ml/kg (超過20公斤部分)
有感流失	55ml/kg	550ml+28ml/kg	830ml+11ml/kg
總量	100ml/kg	1000ml+50ml/kg	1500+20ml/kg
總熱量	100kcal/kg	1000ckal+50kcal/kg	1500kcal+20kcal/kg
鈉	3meq/kg	30mEq+1.5mEq/kg	45mEq+0.6mEq/kg
鉀	2meq/kg	20mEq+1.0mEQ/kg	30mEq+0.4mEq/kg
葡萄糖	5gm/kg	50gm+2.5gm/kg	75gm+1gm/kg

4. Taita Sol'n

①病人血鉀高，無小便，或shock時勿使用

②台大1號+1支50% G/W=5.8% G/W的濃度

台大2號+1支50% G/W=5.3% G/W的濃度 (1支50% G/W在500cc溶液可上升2% G/W)

③1~2歲間台大1號、2號，1瓶500 C.C，台大5號1瓶400 C.C

台大溶液電解質量mEq/L

品名	電解質組成									Glucose g/L	Kcal/L	pH	MOsm/L	ml/ Bot
	Na	K	Mg	Cl	Ca	SO4	Acetate	Gluconate	mM/L Phosphate					
TAITA 1號	25	18	3	20	-	-	20	-	6	38	152	5-6	300	500
TAITA 2號	40	12	-	26	-	-	20	-	6	33	132	5-6	287	500
TAITA 3號	75	12	-	61	-	-	20	-	6	20	80	5-6	285	500
TAITA 4號	110	20	-	102	-	-	16	-	12	8	32	5-6	300	500
TAITA 5號	36	18	3	17	-	-	28	-	12	100	400	5-6	669	400
PAREN-AID 1號	20	24	4	20	5	4	33	5	5	150	600	4-5	946	300
PAREN-AID 2號	30.4	30	8	30.4	-	8	17.5	-	12.5	250	1000	4-6	1517	400
PAREN-AID 3號	30.4	30	8	33	200	8	30	20	-	250	1000	4-6	1557	400

5. Hyperkalemia

①Assessment

a.確認高血鉀及找出高血鉀之因。

b.確認高血鉀引起的症狀及心電圖變化。

②Treatment

a.Mild to moderate(K=6.0-7.0): Goal is enhance excretion of K+

1) Place patient on monitor.
2) Kayexalate (sodium polysterene resin) 1-2 g/kg with 3 ml sorbitol/g resin given PO Q6 hr or with 5 ml sorbitol/g resin as retention enema over 4-6 hours.

b.Severe (K>7.0): Goal is to move K+ into cells acutely.

表一、高血鉀的心電圖變化症狀

血清K(meq/L)	ECG變化
>6.0	T波增高
6.5~7.5	PR間隔及QRS期間長
>8.0	P波消失，QRS-T綜合波變慢，可心室顫動

1) Insulin, regular, 0.1 U/kg IV with 25% glucose as 0.5 g/kg(2 ml/kg) over 30 minutes. May repeat this dose in 30-60 minutes, or begin infusion of $D_{25}W$ 1-2 ml/kg/hr with regular insulin 0.1 U/kg/hr. Monitor glucose hourly.
2) $NaHCO_3$ 1-2 mEq/kg IV given over 5-10 minutes. (May be used in the absence of acidosis.)
3) With onset of arrhythmia, urgent reversal of membrane effects is required. Calcium gluconate (10%) 100mg/kg/dose (1 ml/kg/dose) over 3-5 minutes. May repeat in 10 minutes (dose not lower serum K+ concentration).

4) Dialysis if above measures are unsuccessful.

Note: Ca gluconate solution is not compatible with $NaHCO_3$. Flush lines between infusions.

表二、治療高血鉀症之藥物

藥物	劑量	開始作用時間	藥效持續時間	注意事項
1.Calcium gluconate 10% solution	0.5-1.0 ml/kg	1-5分鐘	1/2-2小時	靜脈注射5-10分鐘並作心電圖監視
2.Sodium bicarbonate 3.75% solution	2.0 mEq/kg	5-10分鐘	2+小時	靜脈注射5-10分鐘
3.Glucose 25%solution and	1-3 ml/kg/hr	1/2-1小時	6-24小時	靜脈注射，速度不可超過0.5 g/kg/hr
Insulin	1 unit/3-5g glucose			必須與25%葡萄糖水同時給予
4.Kayexalate enema	1 gm/kg/dose in 4 ml/kg/10% glucose(可降血鉀1 mEq/l)	1-2小時	4-6小時	灌腸置留1-2小時可以每4—6小時重覆1次
5.透析治療		2^+小時		透析導管可以經皮植入或手術植入

III-1. Fever

- **找Source** :症狀不明顯找不到focus時，尤其應考慮以下診斷:

  1. Infection: AOM (ear drums), Pneumonia (CXR), UTI (U/A,U/C), Abscess(Echo), CNS (神經症狀,CSF), Osteomyelitis (四肢活動), TB(CXR,PPD test), Typhoid Fever(Widal test, Bone Marrow culture), Virus Infection(EBV Ab), Sinusitis
  2. Collagen Disease: Kawasaki disease, SLE …
  3. Malignancy: Leukemia…

- **Fever routine** :

  1. Ice Pillow (<3個月用water pillow) 2. Warm Water Bathing,

  3.藥物使用→參考劑量篇

  4.範例：有差異性,視情況而定 範例：

  >38℃: ice pillow or water pillow

  >38.5℃: 吃退燒藥

  >39℃(吃退燒藥1小時仍>39℃):加用栓劑(拉肚子時儘不給)或Ibuprofen(>6m/o)&溫水浴

III-2. Skin rash 之鑑別診斷

Table 38–3. Red Rashes in Children. (continued)

Condition	Incubation Period (days)	Prodrome	Rash	Laboratory Tests	Comments, Other Diagnostic Features
Drug allergy	—	None, or fever alone, or with myalgia, pruritus	Macular, macuopapular, urticarial, or erythroderma	Leukopenia, eosinophilia	Rash variable. Severe reactions may resemble measles, scarlet fever; Kawasaki disease; marked toxicity possible.
Erythema multiforme	—	Usually none or related to underlying cause	Discrete, red maculopapular lesions; symmetrical, distal, palms and soles; target lesions classic	Normal or eosinophilia	Reaction to drugs (especially sulfonamides), or infectious agents (mycoplasma; herpes simplex virus). Urticaria, arthralgia also seen.
Kawasaki disease	Unknown	Fever, cervical adenopathy, irritability	Polymorphous (may be erythroderma) on trunk and extremities; red palms and soles, conjunctiva, lips, tongue, pharynx. Desquamation is common.	Leukocytosis, thrombocytosis, elevated ESR or C-reactive protein; pyuria; decreased albumin; negative cultures and streptococcal serology; resting tachycardia	Swollen hands, feet; prolonged illness; uveitis; aseptic meningitis; no response to antibiotics. Vasculitis and aneurysms of coronary and other arteries occur (cardiac ultrasound).
Leptospirosis	4–19	Fever (biphasic), myalgia, chills	Variable erythroderma	Leukocytosis; hematuria, proteinuria; hyperbilirubinemia	Conjunctivitis; hepatitis, aseptic meningitis may be seen. Rodent, dog contact.
Staphylococcal scalded skin	Variable	Irritability, absent to low fever	Painful erythroderma, followed in 1–2 d by cracking around eyes, mouth; bullae form with friction (Nikolsky sign)	Normal if only colonized by staphylococci; leukocytosis and sometimes bacteremia if infected	Normal pharynx. Look for focal staphylococcal infection. Usually occurs in infants.
Stevens-Johnson syndrome	—	Pharyngitis, conjunctivitis, fever, malaise	Bullous erythema multiforme; may slough in large areas; hemorrhagic lips; purulent conjunctivitis	Leukocytosis	Classic precipitants are drugs (especially sulfonamides), *Mycoplasma pneumoniae*and herpes simplex infections. Pneumonitis and urethritis also seen.
Toxic shock syndrome	Variable	Fever, myalgia, headache, diarrhea, vomiting	Nontender erythroderma; red eyes, palms, soles, pharynx, lips	Leukocytosis; abnormal liver enzymes and coagulation tests; proteinuria	*Staphylococcus aureus* infection; toxin-mediated multiorgan involvement. Swollen hands, feet. Hypotension or shock.

Table 38–3. Red Rashes in Children.

Condition	Incubation Period (days)	Prodrome	Rash	Laboratory Tests	Comments, Other Diagnostic Features
Enterovirus	2–7	Variable fever, chills, myalgia, sore throat	Usually macular, maculopapular on trunk or palms, soles; vesicles or petechiae also seen	Variable	Varied rashes may resemble those of many other infections. Pharyngeal or hand-foot-mouth vesicles may occur.
Measles	9–14	Cough, rhinitis, conjunctivitis	Maculopapular; face to trunk; lasts 7–10 d; Koplik spots in mouth	Leukopenia	Toxic. Bright red rash becomes confluent, may desquamate. Fever falls after rash appears. Inadequate measles vaccination.
Roseola (exanthem subitum) (HHV-6)	10–14	Fever (3–4 d)	Pink, macular rash occurs at end of febrile period; transient	Normal	Fever often high; disappears when rash develops; child appears well. Usually occurs in children 6 mo–3 y of age. Seizures may complicate.
Rubella	14–21	Usually none	Mild maculopapular; rapid spread face to extremities; gone by day 4	Normal or leukopenia	Postauricular, occipital adenopathy common. Polyarthralgia in some older girls. Mild clinical illness. Inadequate rubella vaccination.
Staphylococcal scarlet fever	1–7	Variable fever	Diffuse erythroderma; resembles streptococcal scarlet fever except eyes may be hyperemic, no "strawberry" tongue, pharynx spared	Leukocytosis is common because of infected focus	Focal infection usually present.
Streptococcal scarlet fever	1–7	Fever, abdominal pain, headache, sore throat	Diffuse erythema, "sandpaper" texture; neck, axillae, inguinal areas; spreads to rest of body; desquamates 7–14 d	Leukocytosis; positive group A*Streptococcus* culture of throat or wound; positive streptococcal antigen test in pharynx	Strawberry tongue, red pharynx with or without exudate. Eyes, perioral and periorbital area, palms, and soles spared. Pastia lines. Cervical adenopathy. Usually occurs in children 2–10 y of age.

III-3. 頸部淋巴腺分布圖

TABLE 17-2. Physiology of Lymph Flow in the Head and Neck

Lymph Node Group	Areas Drained
HEAD	
Postauricular (mastoid)	Temporal and parietal scalp; posterior wall of ear canal; upper half of pinna
Occipital	Posterior scalp; skin of upper, posterior side of neck
Preauricular	Anterior and temporal regions of scalp; anterior ear canal and pinna; lateral conjunctivae
Parotid	Root of nose; eyelids; temporal scalp; exterior auditory meatus; parotid glands; middle ear; floor nasal activity; posterior palate
Facial	Eyelids, conjunctivae; skin and mucous membranes of nose and cheek; nasopharynx
NECK	
Submental	Central lower lip; floor of mouth; skin of chin; tongue tip
Submandibular (submaxillary)	Buccal mucosa; side of nose; medial palpebral commissure; upper lip; lateral part of lower lip; gums, anterior part of tongue margin, teeth
Superficial cervical	Anterior: superficial anterior neck tissues including skin, lower larynx, thyroid, cranial trachea
	Posterior: lower ear canal; parotid region
Superior deep cervical	Tonsil, adenoid; posterior scalp and neck; tongue, larynx, palate; thyroid; nose, nasopharynx; esophagus; paranasal sinuses; all nodes of head and neck except inferior deep cervical
Inferior deep cervical (scalene, supraclavicular)	Dorsal scalp and neck; superficial pectoral region of arm; superior deep cervical nodes; larynx, trachea; thyroid
	Right: left lower lobe, lingula, right lung and pleura
	Left: left upper lobe; entire abdomen

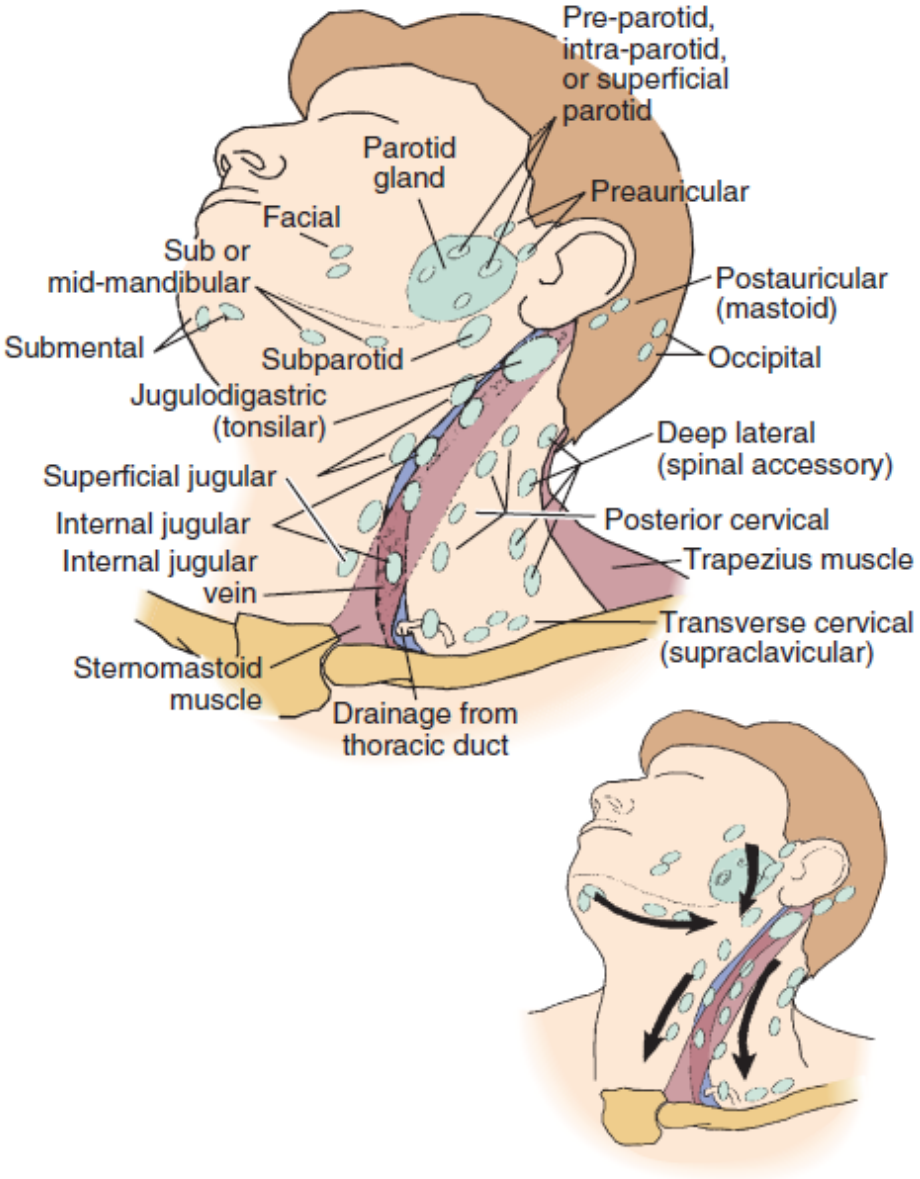


Figure 17-1. Lymphatic flow in the head and neck.

IV. 接各類新病人須知及常用處置方法

1. 新住院病人有**狀況不好者,應優先處理**，並跟其他病床家屬做好溝通。
2. 新病人在診斷及處理上**不了解處,應請教資深醫師**或主治醫師，勿延誤病情。
3. **痰液的檢查:**
   A. **肺炎**或**支氣管炎**
      1) < 2 歲,非細菌性 bronchiolitis 或 pneumonia,可送 sputum RSV Ag (±thoat virus culture)
      2) < 4 個月,非細菌性 bronchiolitis 或 pneumonia,可送 chlamydia Ag 及 RSV Ag
      3) 疑非 RSV 的病毒性肺炎或細支氣管炎,可做 throat virus culture
      4) 疑細菌性肺炎,做 bacterial smear 及 culture 及自費 urine 肺炎球菌檢查
      5) 疑肺結核性肺炎，做 acid-fast stain (AFS) / culture, PPD, gastic, lavage QD×3 天
   B. **化膿性扁桃腺炎**: GAS Ag & culture, virus culture (for adeno virus)，另可抽血 for E-B Ab。(VCA IgG, IgM)
   C. **疑百日咳**: 拿特殊採檢鐵棒，從鼻咽入、咳或停留 15 秒。
5. **新病人疑感染者抽血常規**為 CBC/DC, B/C, CRP, **GOT**,其他則視情形再加抽:
   1) **抽筋**&中樞神經症狀: Na, K, Cl, sugar, BUN, Cre, Ca, (ABG),(NH3)等
   2) **化膿性扁桃腺炎**: EBV antibody (VCA IgG, IgM), ASLO 等
   3) **肺炎超過二歲**:若疑似黴漿菌感染,加抽 Mycoplasma Ab, cold agglutinin test 及 rapid agglutinin test (Bed-side test)
   4) **喘**的小朋友:加抽 ABG or VBG or 生化的 CO_2 及 electrolyte 或疑似心肌炎抽 heart enzyme (troponin-I, CK, CK-MB)等
   5) **腸胃炎及腹疼**: Na, K, Cl, sugar, BUN, Cre (ABG), amylase, lipase, albumin 等
   6) **活力不好**,疑 sepsis: PT, PTT, DIC profile, BUN-Cre, Na, K, Cl, ABG 等
6. **疑 UTI** 的 case,先驗完 **U/C** 後,才給抗生素(若 case 狀況不好，應先治療)
7. **X-ray 1) 哮吼**: ❶chest PA view (含 neck)及 neck, lat. view extended
   2) 肺炎: ❶有 lobar consolidation,照 PA & Lat. view
   ❷疑有肋膜積水: lateral decubitus
   3) 腸炎: ❶KUB (疑 Bacteria infection)
   ❷疑 toxic megacolon,可 q12h 或密集追蹤。
   ❸疑 perforation,照 right side 向上之 lateral decubitus view 或 cross table view, standing view (plain abdomen)。
8. **治療:**
   1) 飲食:
   ❶腸炎:疑 virus, NPO 6 小時再試吃,細菌性腸炎嚴重時 NPO 至少 12 小時。
   ❷呼吸道喘:order 勿開 feeding as tolerated,給予 maintain fluid,先 NPO 或少量 feeding 以不妨礙拍痰為原則。
   2) 吸藥:
   ❶哮吼:給予 cold flow,吸 Bosmin
   ❷氣喘:給予 warm flow,吸 Bricanyl,若很喘可給 A+B,前 6 小時應依狀況吸葯
   ❸肺炎及 bronchiolitis:給 B+J(Jusonine)，效果因人而定，由 VS 個別決定。
9. **查房**:早晚各查一次 data 及巡視病人一次,有狀況時應報告 senior R 或 CR 或 VS。

V. RT consultation

1.開電腦醫囑呼吸治療申請單: 照會呼吸治療師--使用氧氣罩、教導拍痰、吸入藥物、呼吸器設定、裝設 pulse oximeter

2.開 **Admission order**:

1) O_2 Therapy, 2) Chest care (CPT)

①O_2 Therapy:❶ > 5 years old：O_2 mask Nasal cannula

❷ < 5 years old：O_2 hood

❸ O_2 stand by（for p.r.n case）

O_2 hood : Cold flow - 上呼吸道疾患: Croup syndrome

Warm flow (加heater): 1歲前 wheezing infants or hyper-reactive airway

②chest care : RT會教導一次或Stat 請nurse教

- 由於RT只負責教導第一次的拍痰, 其餘均由Family拍痰, 故查房時應隨時**提醒家屬多拍痰(至少2次/天)**，拍痰後請護士評估是否抽痰，**勿一直餵食**(急性喘時牛奶應限正常量一半以下，甚至禁食) 小心咳嗽嗆奶, 變成吸入性肺炎
- 有特殊需要時,可開stat order請R/T幫忙拍痰

3) ***Nebulizer:***

- **Bricanyl (B), Jusomine (J),** N-Acetylcysteine (Fluimucil, N-AC), **Atrovent (A)** or **Bosmine** inhalation, **Stat and PRN,** 有開Stat order才會馬上吸藥, 如果開立 routine order則是 Tid (9 am - 3 pm - 9 pm)。

①吸藥 Nebulizer：依**嚴重度**決定(如需密集吸入，請確實評估病人的必要性，是否轉ICU)，吸藥間歇依喘或痰嚴重度而定。

1.下呼吸道：

- Pneumonia時 J (化痰) ± B
- Bronchiolitis 時B + J (化痰)，但效果依人而異，請依VS的order而定
- Asthma時: B ± A(大喘)

Asthma acute attack大喘時，前1小時可20分鐘吸1次A+B

2.上呼吸道 (Croup)：吸Bosmin, 很喘前1小時可20分鐘吸一次, 一般約吸3天。

VI. MODIFIED GLASGOW COMA SCALE

E, Eye-opening Response(睜眼反應)	
Spontaneous(主動地睜開眼睛)	4
To speech(聽到呼喚後會睜眼)	3
To pain(有刺激或痛楚會睜眼)	2
None(睜不開)	1
V, Verbal Response(語言反應)	
Oriented(說話有條理，會與人交談)	5
Confused(可應答，但說話沒有邏輯性)	4
Inappropriate words(可說出單字或胡言亂語)	3
Incomprehensible sounds(可發出呻吟)	2
None(無任何反應)	1
T(氣切), E(氣管插管),A(失語症, aphasia)	
M, Motor Response(動作反應)	
Obeys(可依指令做出各種動作)	6
Localizes(施以刺激時，可定位出疼痛位置)	5
Withdraws(對疼痛刺激有反應，肢體會閃避)	4
Decorticate flexion(去皮質, 對疼痛刺激肢體會彎曲，試圖迴避)	3
Decerebrate extension(去大腦, 對疼痛刺激肢體反而會伸展開)	2
None(無任何反應)	1
Verbal Response for Young Children and Adults	
Appropriate words or social smiles, fixes and follows	5
Cries, not consolable	4
Persistently irritable	3
Restless, agitated	2
None	1

昏迷程度以 E、V、M 三者分數加總來評估, 正常人昏迷指數最高 15 分,最低 3 分, 若因插管或氣切無法發聲的重度昏迷者會有 2E(氣管插管)或 2T(氣切) 的評分

輕度昏迷: 13-14 分

中度昏迷: 9-12 分

重度昏迷: 3-8 分

This is not uniformly accepted as the definitive grading system for children.

Modified from Luerssen TG: Acute traumatic cerebral injuries. In Check WR (eds): Pediatric Neurosurgery, 3rd ed. Philadelphia. WB Saunders, 1994, p.2

VII. Oncologic Emergency

Spinal Cord Compression

- Compression for more than 24 hours may produce irreversible neurological damage

Etiology:

- Intradural lesions (primary central nervous tissue tumors, drop metastasis from medulloblastoma, ependymoma, astrocytoma)
- Direct extension of paravertebral tumors (dumbbell lesions: neuroblastoma, Ewing, rhabdomyosarcoma)
- Prednisolone, methotrexate → osteoporosis with vertebral body collapse
- LP in thrombocytopenic patient resulting in hematoma

Sign and symptoms:

- Pain (central back pain, vague deep ache, radiating pain to groin or leg, tenderness, irritability in infants)
- Weakness (usually proximal)
- Sensory changes (hyperesthetic band above sensory level)
- Bowel and bladder involvement (generally late findings, constipation and urinary retention precede incontinence)

Diagnosis:

- Careful physical examination (strength, gait, sensory exam., deep tendon reflexes, superficial reflexes, anal tone, cranial nerves)
- CXR, spinal films
- CT scan
- MRI with gandolinum
- Tissue diagnosis (lymph node biopsy, BM, laminectomy specimen)
- LP ***contraindicated***

Treatment:

- Immobilize patient
- Pain relief, sedation
- Steroid (dexamethasone)
- Radiation (for multiple level lesions, advanced metastatic disease, bulky unresectable tumors)
- Chemotherapy (for responsive tumors such as lymphoma, neuroblastoma, Ewing sarcoma, rhabdomyosarcoma)
- Surgical decompression indicated
  a. To establish histological diagnosis
  b. For high cervical blocks with respiratory compromise
  c. Failure of chemoradiotherapy
  d. Stabilization of vertebral body collapse to relieve pain
- Physical therapy
- Stool softeners
- Bladder drainage
- Close monitoring, daily neurological examination

Tumor Lysis Syndrome

Etiology:

- Occurs in patients with a large cell burden of tumor exquisitely sensitive to chemotherapeutic agents (Burkitt's lymphoma, T-cell ALL with high white count)
- It is a consequence of rapid release of intracellular metabolites
  a. uric acid

b. potassium
c. phosphate

Hyperuricemia:

- Results from degradation of purine released by fragmented tumor nuclei
- The progressive precipitation of uric acid crystals in the distal tubules may produce obstructive uropathy, oliguria, azotemia

Treatment:

- IV hydration 3000 ml/m^2/24 hr → extracellular volume expansion to increase urate excretion
- NaHCO3 150-210 mEq/M^2/24 hrs (add 50-75 mEq/L and adjust to keep urine pH > 7) → alkalization of urine to maximize urate solubility
- Allopurinol 10 mg/kg/day divided Tid → decrease urate production by competitive inhibition of xanthine oxidase; Allopurinol is rapidly absorbed after oral administration (assuming normal gastrointestinal function), peak levels being reached within 2 hours of administration
- Acetazolamide 150 mg/M^2 q6-8 h may help alkalinization of urine

Hyperkalemia

- Results from acute tumor lysis and may worsen with developing acute renal failure secondary to hyperuricemia or hyperphosphatemia
- EKG - **tented T waves, wide QRS**
- Treatment:
  1. kayexalate 1-2 g/kg in 20% sorbitol per rectum
  2. Ca. Gluconate 10% 0.5 ml/kg over 10 min. (not in same line with NaHCO3 → may precipitate
  3. NaHCO3 1-2 mEq/kg over 5-10 min
  4. glucose 0.5 g/kg iv with RI ***0.25 U/g*** of glucose iv over 30 min.

Hyperphosphatemia and hypocalcemia

- Results from release of phosphate from tumor cells (lymphoblasts have a phosphate concentration that is **4-fold** higher than normal lymphocytes)
- Secondary to ARF from precipitation of uric acid
- When **the product of PO4 x Ca > 60**, calcium phosphate crystals precipitate in the microvasculature and renal tubules resulting in hypocalcemia and tetany
- Treatment :
  1. hydration
  2. Ca. gluconate 200-500 mg/kg/24 h continuous infusion
     Caution : because increase of Ca x PO4 product will increase precipitation of CaPO4 crystals, **Ca. gluconate should only be started when serum calcium is very low. i.e. < 7 mg/dl and symptoms of tetany are present**
  3. DC alkalization ,∵ **CaPO4 crystals are less soluble in alkaline pH;** alkalosis will decrease ionized Ca^{2+} leading to tetany
  4. Aluminum hydroxide (Alugel) 100-150 mg/kg/day Tid (slow action)
  5. If above fail consider dialysis

Hypercalcemia

Etiology:

- Infrequent in childhood malignancies, may be seen with leukemia, lymphoma or solid tumors. **Although several cytokines such as interleukin 1 and parathyroid hormone-related protein are thought to be involved in the development of high serum Ca(++) levels, its precise underlying mechanism remains unknown.**
- Increased local release of skeletal calcium caused by direct erosion of bone

- **Release of local or systemic endocrine factors**

Symptoms and signs
- Nausea, vomiting
- Polydipsia, polyuria
- Constipation, abdominal pain
- Lethargy, irritability
- Hypertension, arrhythemia, seizures

Treatment :
1. *i.v.* hydration 3000 ml/M^2/24 h with normal saline (extracellular volume expansion to increase Ca. excretion)
2. **Lasix 1-2 mg/kg iv q6h (block reaborption of Ca. in ascending loop of Henle)**
3. Calcitonin 4-8 IU/kg SC or IM q12h (short-term inhibition of osteoclastic bone resorption; renal calciuretic effect)
4. Prednisone 2 mg/kg/day (may enhance effect of calcitonin)
5. Mithramycin 25 *u*g/kg (1-3 daily doses) max. weekly dose 150 *u*g/kg (inhibition of osteoclastic bone resorption; toxic to bone marrow, liver and kidney)

Superior Vena Cava Syndrome

Etiology:
- Vascular compression caused by an anterior superior mediastinal mass (lymphoma, germ cell tumor, neuroblastoma)
- SVC obstruction (thrombosed indwelling central venous catheters)
- Superior mediastinal syndrome (tracheal or bronchial compression by an anterior superior mediastinal mass

Signs and symptoms:
- Distention of neck veins, edema above the clavicles, facial plethora
- Respiratory distress (cough, retractions, stridor, wheezing, cyanosis)
- Supine position usually worsens condition
- Pleural effusion

Diagnosis:
- Minimal invasive approach to diagnosis
- CBC, D/C, BM, peripheral lymph node biopsy, thoracentesis
- CXR, CT, (echo)

Treatment:
- When symptoms not life threatening - biopsy and start specific treatment
- With life threatening symptoms - elevate patient's head, O2, avoid sedative, intubation may be very difficult, steroids (dexamethasone), emergency radiation and/or chemotherapy, establish correct diagnosis as soon as safely possible
- Thrombosis from indwelling catheter - remove catheter, dissolve or surgically remove clot
- **Venous access through lower extremity**, iv fluid avoid overhydration
- Allopurinol, alkalinization of urine

VIII. 抽搐

Basic Concepts

1. What is a seizure?

1) It is a temporary alteration of brain function cause by paroxysmal, abnormal cerebral neuronal discharge.
2) 在臨床上造成意識障礙或昏迷，動作上、行為上或情緒上的異常，感覺上或自主神經系統上的功能障礙。
3) 抽搐發作，有時只有異常的動作發生，並不合併意識的喪失。

2. What is a convulsion?

A seizure causing abnormal muscle contractions.

3. What is epilepsy?

1) Epilepsy 是由許多先天或後天因素所引起之慢性腦疾病，是由於腦細胞之過度放電所引起的反覆性發作，而伴隨著各種不同的陣發性臨床症狀。
2) 是由於腦障礙而引起之慢性疾病過程。
3) 症狀出現之特點為間歇性(intermittent)、陣發性(paroxysmal)、短暫性(transient)、反覆性(repetitive)、及不定時再發性(recurrent)。

PE & NE

- Special skin lesion or facial dysmorphism?
- Macrocephaly or IICP sign (papilledema, buldging anterior fontanelle, bradycardia, hypertension)?
- Dehydration or meningeal sign?
- Abnormal bruit heard in skull or periorbital region?
- Twelve pairs cranial nerves, muscle power, DTR, muscle tone, cerebellar sign, JOMAC, coma scale...

Clinical Test

EEG(sleep, sleep & awake, awake, video EEG), brain image study(brain echo, CT, MRI), lumbar puncture.

Diagnosis

History

- Any underlying problems (limitation in motor function, mental retardation, failure to thrive, or special syndrome)?
- Any insult before (encephalitis, hypoxic-ischemic encephalopathy)?
- Taking anticonvulsant or not?
- Changing dose or type of anticonvulsant drug?
- Drug compliance?
- Recent infection or not?

- Change in seizure type or duration?
- Delay developmental milestone?

Pre-ictal symptoms?

Ictal symptoms?

Post-ictal symptoms?

Management

1. Seizure control

- Stabilize vital sign- O2
- Valium : 0.2 - 0.3mg/kg.dose iv or 0.5 mg/kg.dose per-anus
- Midazolam: 0.15mg/kg.dose iv or im
- Ativan: 0.1mg/kg.dose slowly iv, 0.05mg/kg.dose bolus 10 mins later prn

Consider start anticonvulsants if frequent seizure attacks:

- **In patient who never take anticonvulsant before:**
  - Phenobarbital (luminal): 10-20 mg/kg. dose q12 hrs twice loading, then 2.5 mg/kg.dose q12hrs maintenance
  - Phenytoin (dilantin): 10-20 mg/kg.dose q12 hrs twice loading, then 2.5 mg/kg.dose q12hrs maintenance
- **In patient who is taking anticonvulsant(s) at present:**
  - Check drug level (trough or at attack)
  - Give IV form anticonvulsants (luminal and dilantin) in poor controlled patient.

    a. If the patient is taking luminal, dilantin or LA at present: give boost dose 5mg/kg.dose.

    b. If the patient is not taking luminal, dilantin or LA at present: give loading and maintaince dose as above.
  - Arrange EEG (sleep only if age < 5 years old; order chloral hydrate(10%) 0.5ml/kg po for sedation; maximal 30 ml/dose).
  - Do not stop previous anticonvulsants. If NPO including oral medication is necessary, make sure IV form anticonvulsant(s) have be given already.

IX. 急性氣喘發作的評估及處理

1. 嚴重氣喘發作的危險因子

Infancy
History of severe disease, eg, respiratory failure, hypoxic seizures, frequent nocturnal symptoms, rapidly progressive attacks, prior need for repeated oral steroid courses
Prior hospital or intensive care unit admissions
History of poor compliance
Frequent beta-2 agonist use
Labile peak expiratory flow rates (PEFR)
Prolonged duration of recent acute symptoms
Delayed or no usage of oral steroids during attack
Inadequate social support system
Depression or manipulative use of asthma
Hyperinflation and increased lung compliance
Severe atopic disease

2.何時需安排 CXR 檢查

- Physical signs of an extra-alveolar air leak (pneumomediastinum, pneumothorax, subcutaneous emphysema)
- Chest pain
- Fever ≧38.9℃ (102℉)
- Rales
- Following intubation to assess endotracheal tube placement

3.氣喘急速惡化的嚴重度

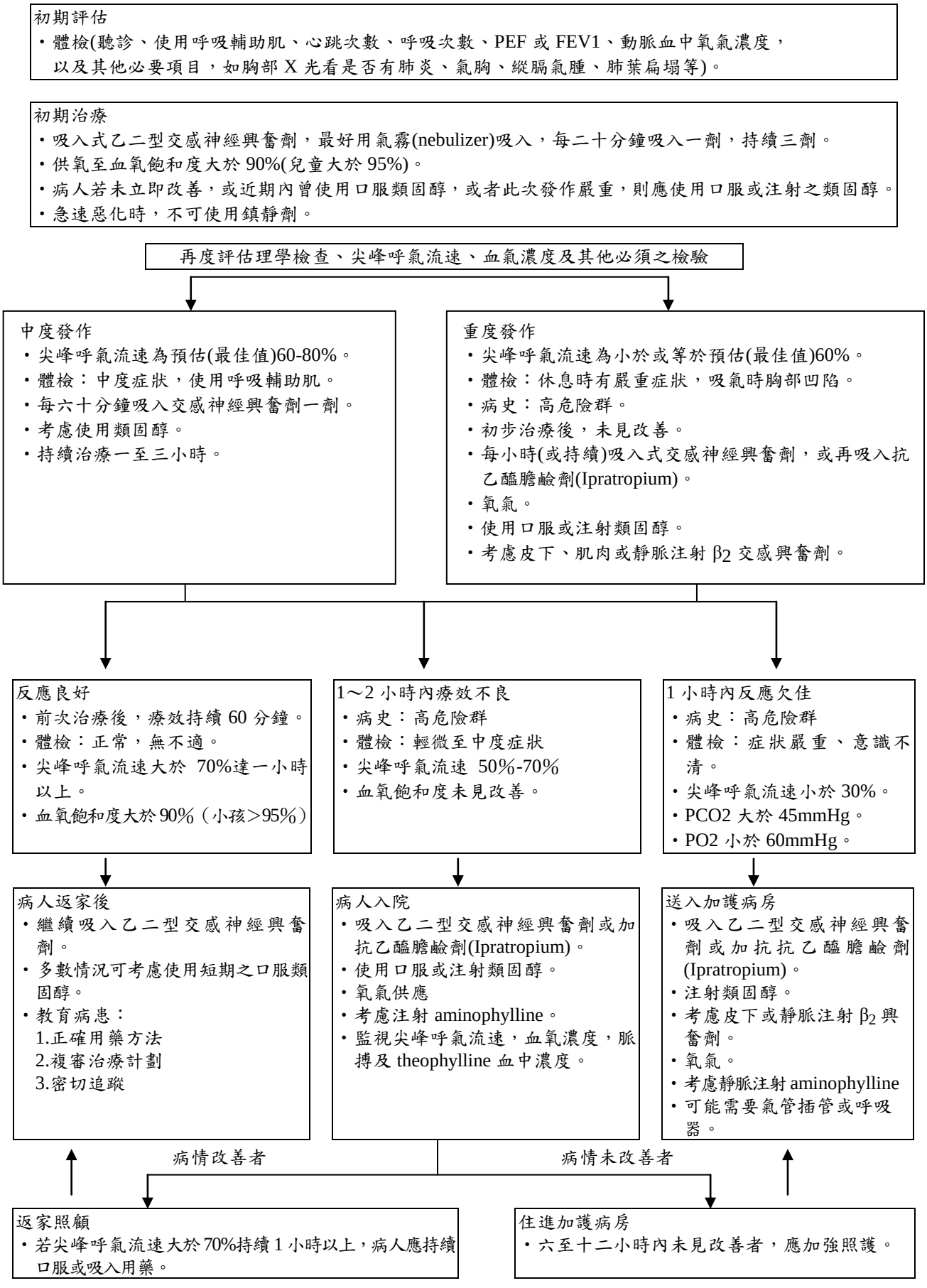
	輕度	中度	重度	呼吸衰竭;緊急
喘息程度	走路 可以躺下來 嬰兒-哭聲短 弱，餵食困難	說話 喜歡坐著	休息 向前彎腰	
說話長度	句子	片語	單字	
意識狀態	可能焦躁	通常焦躁	通常焦躁	嗜睡或意識不清
呼吸速率	增加	增加	>30 次/分	
使用呼吸輔助肌，胸骨上方凹陷	通常沒有	通常有	通常有	胸腹反常運動
喘鳴聲	中度，通常在呼氣末期	大聲	通常大聲	聽不到
心跳（數/分）*須考慮年紀	<100	100-120	>120	心跳變慢
開始支氣管擴張劑治療後的尖峰呼氣流速(預估值或最佳值的百分比)	>80%	60-80%	<60%(成人<100公升/分)或支氣管擴張劑療效維持不到 2 小時	
動脈血氧分壓	正常 通常不需檢查	>60mmHg	<60mmHg 可能發紺	
動脈血二氧化碳分壓	<45mmHg	<45mmHg	>45mmHg 可能呼吸衰竭	
動脈血氧飽和度	>95%	90-95%	<90%	

幼兒比成人及少年易發生缺氧和二氧化碳分壓升高(換氣不足)，在嬰兒與小孩當脈衝式測氧機顯示 SaO_2 低於 95%時需做動脈血氣體分析。

兒童正常脈搏	兒童正常呼吸速率
2-12 月 ＜160/分	＜ 2 月 ＜60/分
1- 2 歲 ＜120/分	2-12 月 ＜50/分
3- 8 歲 ＜110/分	1- 5 月 ＜40/分
	6- 8 歲 ＜30/分

只要出現數個項目(不需要全部出現)，便能夠適用上述氣喘發作嚴重度之分類。

4.氣喘急速惡化的處理：醫院的處置



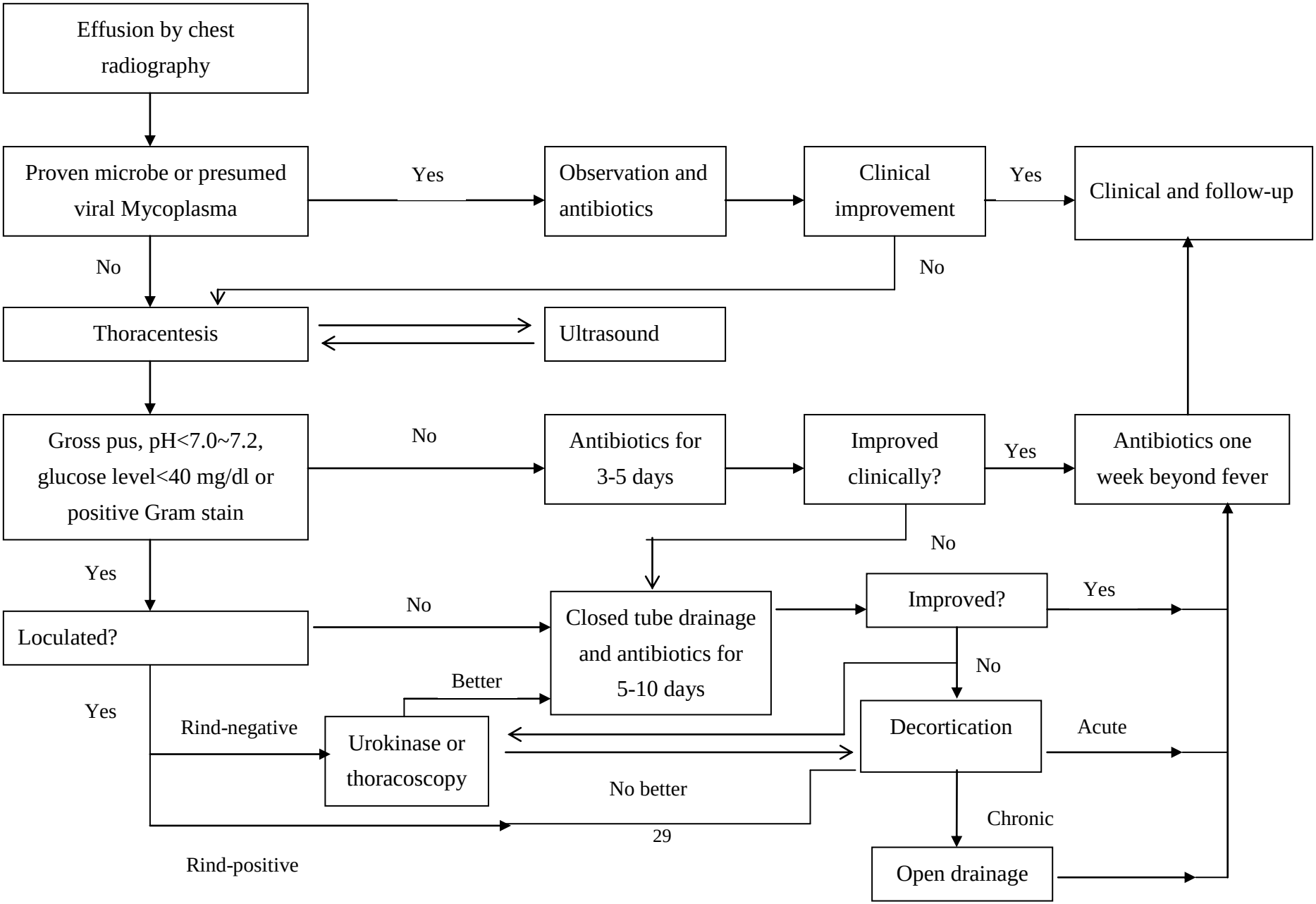
5.急診住院的標準

◎ Patients having a good response to therapy usually can be discharged to home with an appropriate follow-up plan. "Good" response has been defined by the National Asthma Education Program Expert Panel as a PEFR >70 percent of the patient's baseline, an oxygen saturation >95 percent, and improvement in other clinical parameters.

◎ An incomplete response to initial therapy (PEFR 40 to 70 percent of patient's baseline, oxygen saturation 91 to 95 percent, and improving but not optimal physical signs) should prompt continued treatment. This may be carried out in an observation unit if one is available adjacent to the emergency department or else on the hospital ward.

◎ A poor response to therapy is indicated by a PEFR <40 percent of baseline, an oxygen saturation <91 percent, and no improvement in clinical features of airflow obstruction. Usually this pattern is apparent after 1 to 2 hours of intensive emergency department treatment and is unlikely to abate within 24 hours. Hospital admission should be prompt for such patients.

X. Algoritlun for the management of pleural fluid collections



XI-1. 緊急肋膜穿刺術之一

Definition/purpose

Needle tap into pleural space for emergency removal of air or fluid.

Technique

1.Position patient in supine or semi-fowlers.

2.Locate area to be tapped by physical exam, transillumination, or x-ray (or echo)

Most common sites:
Air—2nd –3rd IC, midclavicular line or anterior axillary line Fluid—5-6 IC, mid-axillary line.

3.Clean skin with alcohol and/or iodine.

4.Attach 18 gauge needle(child) or 21 gauge scalp vein needle (infant) to a 3-way stopcock and 20 ml syringe.

5.A flexible cannula may be used to replace the needle once the puncture is made.

6.Insert needle just over superior margin of lower rib at 90° angle.

7.Once needle is under the skin, have assistant apply gentle negative pressure to syringe plunger.

8.Advance needle, usually a decrease in resistance and a flow of air into the syringe occurs as the pleura is punctured.

9.Hold needle as still as possible.

10.When syringe is full, turn off stopcock, remove syringe and empty it.

11.Reconnect syringe to stopcock and repeat until air flow stops.

12.Close monitoring of vital signs is essential during procedure.

13.Remove needle and cover site with petroleum gauze and small dressing.

14.Obtain stat chest x-ray.

15.This procedure should be followed in most cases by a permanent chest tube.

N **O** **T** **E**	Caution:Needle should be removed if patient coughs repeatedly. Do not leave needle in place when procedure is finished

Complications

Pneumothorax Hemothorax Infection

XI-2. Thracentesis (肋膜穿刺術)之二

1.和家屬解釋 thracentesis 的目的、方法、危險性，請家屬簽同意書。

2.找一位孔武有力的醫護人員，將病童面對面用力抱緊(像抱無尾熊)，坐在椅子上，若病童太躁動，可使用 sedation 但請注意呼吸。

3.確定 tapping 位置，千萬不要搞錯左右，(可 echo-guided)病人身體可傾向有水那邊。

4.定位：個人喜好從 mid-axillary line 和 5th ICS 上緣交接處下針。(此為高位，定位應綜合 CXR、理學檢查及 echo)

5.使用沖洗棉枝，iodine/Alcohol，消毒三套，面積大一些。

6.無菌操作：戴手套，取 18#IC 針，T 延長管(或 3 way stopcock)及 20ml 空針，以手套內包裝作無菌面，組合 T 延長管及 20ml 空針，將 IC 針及組合物置於無菌面上。

7.像 set IV 的技術一樣，將 18# IC 針刺入定位點，肋骨上緣下針，等到 pressure release 或看到有回水，固定硬針，推入軟針。

8.拔出硬針，用手指按住 IC 軟針開口，避免 air 跑進去。

9.將組合完成的 T 延長管(或 3 way stopcock)及空針，接上 IC 軟針。

10.抽水，儘量抽。

若抽不到水，有可能是 IC 軟針 kinking 或 IC 軟針 tip 碰到肺組織，此時可將病童傾至 depending site，或前後移動 IC 軟針。

11.抽出來的 pleural effusion 送檢測(case by case)PH(同 ABG 方式)、routine、gram stain、rapid Ag、cytology, sugar、LDH、total protein、culture(Bac, virus, fungus, TB if indicated)，若 effusion 檢體不足，請送 culture 及 gram stain。

12.抽完拔出 IC 軟針，乾紗 cover 洞口，外蓋布膠。

13.照胸部 X 光，確認無氣胸。

XII. CHEST TUBE INSERTION

Indication

Evacuation of pneumothorax or large pleural-fluid collection.

Equipment

- Pediatric Cut-Down Set Tray
- Thoracic Catheter
- Betadine Solution
- Sterile Gown And Gloves
- Cap And Mask
- Waterseal Drainage Unit
- 3-0 Silk Suture
- Cardiorespiratory Monitor

Chest Tube Sizes(Fr)	
Newborn	8-12
Infants	14-20
1-5yr	20-24
5-10yr	24-30
>10yr	26-32
>16yr	28-42

Technique

1. Assemble equipment(prepare drainage unit and open trays).
2. Position and restrain patient(insure NTE if infant)
3. Insure continuous cardiorespiratory monitoring.
4. Use sterile surgical technique(gown, glove, cap and mask).
5. Clamp distal end of catheter.
6. Pinch proximal end of catheter with curved hemostat.
7. Prepare insertion site(Betadine and drape)(Do not block heat source)

Emergency(not x-ray confirmed)-2nd IC space, midclavcular line
X-ray confirmed-4th or 5th IC, anterior axillary line

1. Make superficial incision(1 cm long, upper border of lower rib).
2. Separate muscle tissue(use small straight hemostat).
3. Puncture the pleura with the curved hemostat(and catheter)(push downward and away from lung tissue).
4. Slowly release and remove hemostat.
5. Slowly advance catheter to desired length(at least 3 cm beyond last opening in catheter).
6. Connect distal end of catheter to water seal drainage unit and turn on low suction(60-80mmHg).
7. Remove distal end clamp(insure all connections are secure).
8. Secure catheter with purse-string suture.
9. Apply occlusive dressing(vaseline gauze and sterile dressing).
10. Obtain x-ray. document.

Care of Chest Tube	Removal of Chest Tube
Secure tubing to bed to avoid tension. Mild the tubing q 1 hr.(for fluid) Observe for changes in air flow or fluid vol or color. Monitor patient's vital signs. Watch for resp. distress, cyanosis, or HR changes. Transport of patient: Clamp tube with Hemostat if drainage system will be elevated above insertion site. Use portable suction system if tension pneumothorax is present. Never clamp a chest tube without an order.	Indications-no bubbling for 24 hrs. no pneumothorax on x-ray. Technique-clamp tube or H2O seal Wait 12-24 hrs. Obtain chest x-ray. If no reaccumulation has occurred, then remove tube. Remove tube during expiration-draw the ends of the purse-string suture together. Immediately cover Puncture hole with Petroleum jelly gauze And plain gauze. Apply sterile dressing. Monitor vital signs and obtain x-ray in 1-2 hrs.

Complications to watch for

- Perforation of lung or major vessels (pneumo/hemothorax).
- Nerve damage-intercostals or phrenic.
- Tube misplacement
  - Tip impinging on aorta will cause hypotension.
  - Placement too low may injure liver, spleen or kidney.
  - If most proximal fenestration is not inside pleural cavity, subcutaneous emphysema may result.
  - Improper placement may fail to relieve tension pneumothorax.
- Equipment malfunction
  - Leaks
  - Blockages
  - Incorrect suction pressures
- Infection (empyema)

XIII. 置放肛管注意事項

(一)材質：

1.軟硬適中

2.Size 適中，一般 7 號以上 Nelaton 可用來 decompression，可選擇約與病人小指同粗之 Nelaton 或單導尿管。

(二)用途

1.Decompression

2.Irrigation

(三)準備事項

1.手套

2.jelly

3.Warm normal saline-irrigation 用

(四)Indication

1.Megacolon-for irrigation 及 decompression

2.Colitis with toxic megacolon

3.Colon preparation-術前準備

4.Nelaton 前端剪數個洞以利 drainage

(五)Contriaindication

近日內接受過肛門手術，如 Imperforate anus 接受 PSARP 者，或 Megacolon 接受過 pull through operation。

(六)注意事項

1.要放置 rectal tube 前請先仔細判讀 plain abd x-ray，如果 bowel obstruction level 在 Small bowel 者，放置肛管並沒有作用。

2.放 Rectal tube 前，請先執行 digital examination 後再放入，避免在 colon 內 twist，必要時可 follow up x-ray。

3.放 Rectal tube 的深度須依 bowel dilatation 的範圍決定，如 Sigmoid dilatation 而 Rectal collapse 時，肛管須放到 dilatation 的 colon 才能發揮 decompression 效果。

4.Colitis 時，x-ray 可見到 severe colon dilatation，此時 mucosa 上會有許多 ulcer，最好先照會外科醫師來看，如需 on rectal tube decompression 時，一定要很 gentle 下進行。

5.Megacolon with severe colonic obstruction 時，先作 digital examination 後，如有大量 stool，foul gas rush out 時，前述的診斷即可能成立，可立即放置 Rectal tube for decompression 及用 warm normal saline irrigation，Abdominal distension subsided 後再安排後面的 LGI，Suction biopsy 及 mannometrium Study。

XIV. 新生兒

XIV-1. Apgar 計分法

徵候	0	1	2
心跳	無	每分鐘 100 以下	每分鐘 100 以上
呼吸力道	無	慢而不規則	有力的哭泣
肌肉張力	癱軟無力	四肢稍微屈曲	有力的活動
反射	無反應	作鬼臉狀	咳嗽或打噴嚏
膚色	藍色、蒼白	身體粉紅，四肢發紺	全身呈粉紅色

XIV-2. 氣管內管的選擇

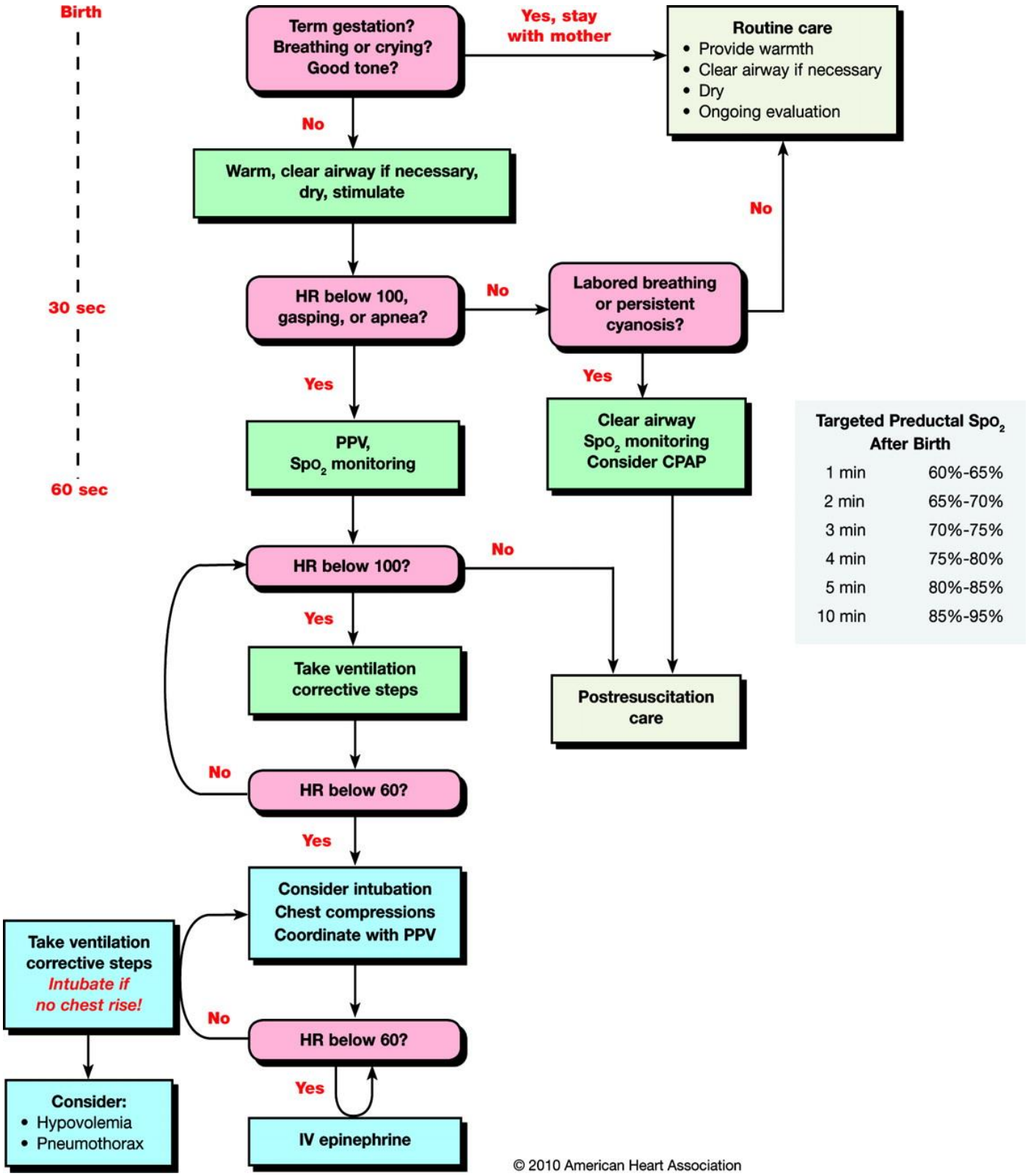
氣管內管內徑(公厘)	體重(公克)	放置深度
2.5	低於 1000	7cm
3.0	1000~2000	8cm
3.5	2000~4000	9cm
3.5~4.0	大於 4000	10cm

XIV-3. 新生兒之抗生素劑量

		劑量(mg/kg/day)與給藥間隔			
		體重≤2000g		體重≥2000g	
抗生素	給藥途徑	年齡 0-7 天	>7 天	年齡 0-7 天	>7 天
Amikacin	IV,IM	15 q12h	22.5 q8h	20 q12h	30 q8h
Ampicillin	IV,IM				
Meningitis		200 q12h	300 q8h	300 q8h	400 q6h
Other diseases		100 q12h	150 q8h	150 q8h	200 q6h
Aztreonam	IV,IM	60 q12h	90 q8h	90 q8h	120 q6h
Cefotaxime	IV,IM	100 q12h	150 q8h	100 q12h	150 q8h
Ceftazidime	IV,IM	100 q12h	150 q8h	150 q8h	150 q8h
Ceftriaxone	IV,IM	50 q24h	50 q24h	50 q24h	75 q24h
Clindamycin	IV,IM,PO	10 q12h	15 q8h	15 q8h	20 q6h
Erythromycin	PO	20 q12h	30 q8h	20 q12h	30 q8h
Gentamicin	IV,IM	5.0 q12h	7.5 q8h	5.0 q12h	7.5 q8h
Meropenem	IV	40 q12h	40 q12h	40 q12h	60 q8h
Imipenem	IV	40 q12h	40 q12h	40 q12h	60 q8h
Metronidazole	IV,PO	7.5 q24h	15 q12h	15 q12h	30 q12h
Oxacillin	IV,IM	50 q12h	75 q8h	50 q12h	1~200 q6h
Penicillin	IV				
Meningitis		200,000U q12h	300,000U q8h	300,000U q8h	400,000U q6h
Other diseases		100,000U q12h	150,000U q8h	150,000U q8h	200,000U q6h
Penicillin	IM				
Benzathine		50,000U	50,000U	50,000U	50,000U
Procaine		(one dose)	(one dose)	(one dose)	(one dose)
		50,000U q24h	50,000U q24h	50,000U q24h	50,000U q24h
Ticarcillin	IV,IM	150 q12h	150 q12h	150 q12h	225 q8h
Tobramycin	IV,IM	3 q24h	4 q18h	5 q12h	7.5 q8h
Teicoplanin	IV	8 qd	8 qd	8 qd	8 qd
Vancomycin	IV	30 q12h	45 q8h	30 q12h	45 q8h

XIV-4. 2010 年新生兒急救復甦流程

NALS Algorithm for Neonatal Resuscitation



Reference: Circulation 2010;122:S909-19.

XIV-5. Neonatal Hyperbilirubinemia

Evaluation and management:

1. blood samplings: bilirubin T/D, CBC/DC, reticulocytes, U/A, U/C, Urine GBS, B/C, 懷疑 infection 要做 sepsis workup.
2. Advanced study: blood typing, Rh minor group, direct Coombs' test, PB smear, G6PD activity, VDRL, HBs Ag, TORCH IgM, thyroid function
3. history: appetite, activity, breast feeding, blood typing, age of onset, hereditary disease, stool color…etc.
4. physical exam: extent of jaundice, cephalohematoma, bruise, **signs of bilirubin encephalopathy**
5. Special considerations: IV for dehydration, antibiotics for bacterial infection, blood smear for spherocytosis, abdominal echo for adrenal hematoma….

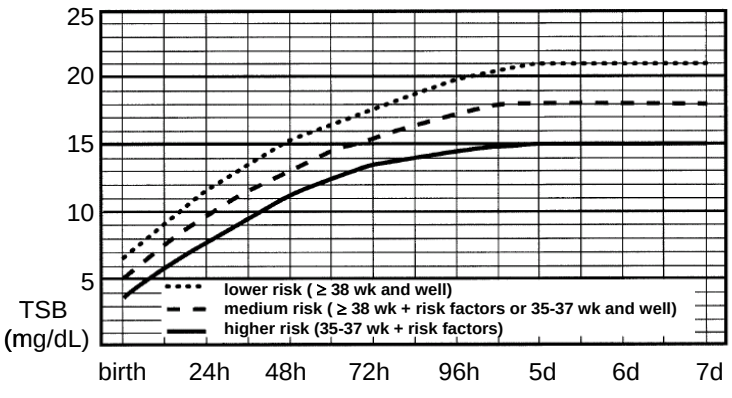
Guideline for preterm infants <u>less than 35 weeks</u> of gestational age[a]

	Total Serum Bilirubin (TSB) (mg/dL)	
birth weight (g)	**phototherapy**	**exchange transfusion**
< 1500	5 – 8	13 – 16
1500 – 1999	8 – 12	16 – 18
2000 – 2500	11 – 14	18 – 20

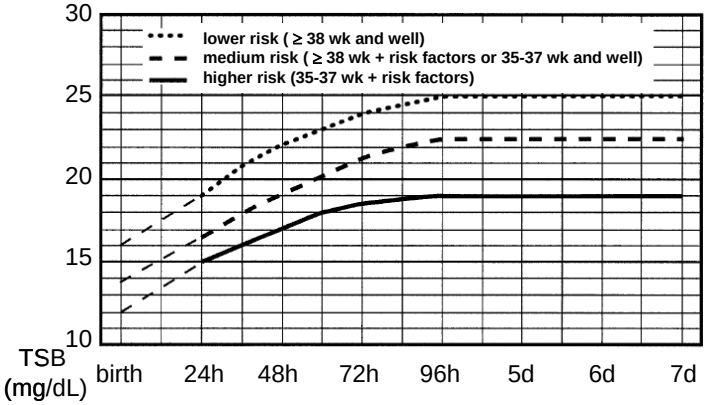
Risk factors:

isoimmune hemolytic anemia, G6PD-deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis, albumin < 3.0 g/dL

Guideline for phototherapy[b]



Guideline for exchange transfusion[b]



References:
a. Arch Dis Child Fetal Neonatol Ed 2003; 88: 459-63
b. Pediatrics 2004; 114: 297-316

XIV-6. Newborn Physical Examination

The newborn infant should be examined within 24 hours after birth.

I. Vital signs

A. Temperature
B. Respirations: normal 40~60/min
C. Blood pressure
D. Pulse rate: normal 100~180/min, usually as low as 70~80 while sleeping

II. Head circumference, length, weight, and gestational age

E. Head circumference and percentile: normal 32~37 cm at term
F. Length, percentile
G. Weight, percentile
H. Assessment of gestational age (see appendix)

III. General appearance: activity, skin color, congenital abnormalities

IV. Skin

A.Color: jaundice, pallor, and cyanosis, bruise, Harlequin coloration, and mottling should be noted

Cyanosis:

a. ***Central cyanosis:*** IT may be associated with congenital heart or lung disease.

b. ***Peripheral cyanosis***: Bluish skin with pink tongue and lips. Peripheral cyanosis may be associated with met-Hb.

c. ***Acrocyanosis***: Bluish extremities. Acrocyanosis may be normal for any newborn that has just born or one experience cold stress.

B. Hereditary vascular lesion: most hemangioma will regress later in the life.

Port-wine stain: If the lesion appear over the forehead and upper lip, then Sturge-Weber syndrome needs to be ruled out.

V. Head:

A. Frontanelles
B. Molding
C. Caput succedaneum, Cephalohematoma, Subgaleal hemorrhage should be differentiated.
D. Increased Intracranial pressure: The following can be seen: bulging frontanelles, separated sutures, paralysis of upward gaze (setting-sun sign), and prominent veins of the scalp.
E. Craniosynostosis: Premature closure of one or more sutures of the skull.
F. Craniotabes

VI. Neck: Palpate for any mass noted. A short neck is seen in Turner syndrome and Noonan syndrome.

VII. Face: Facial asymmetry with crying may indicate facial nerve paralysis or oris ocularis hypoplasia.

VIII. Ears: Look for unusual shape or abnormal position. Low-set ear and fistula should be noted.

IX. Eyes: light reflex, hypertelorism, pupil, movement tracking

X. Nose: Verify the patency of the nostrils with an NG tube if choanal atresia is suspected. Nasal

flaring is indicative of respiratory distress.

XI. Mouth: Cleft palate, Ranula, Epstein's pearls, macroglossia should be noted.

A.Macroglossia: Hypothyroidism, Beckwith syndrome or Pompe disease

B.Frothy or copious saliva: T-E fistula

XII. Chest:

A.Observation: An asymmetric chest may signify a tension pneumothorax. Tachypnea, sternal and intercostals retraction, and grunting on expiration indicate respiratory distress.

B.Breath sounds: Absent or unequal breath sounds may indicate pneumothorax or atelectasis.

C.Breasts may be abnormally enlarged (unilateral or bilateral) secondary to the effects of maternal estrogens or mastitis.

XIII. Heart:

Listen for murmurs: record the pattern, grade, and location of murmur

XIV. Abdomen

A.Observation: Obvious defects may include omphalocele or a gastroschisis. A scaphoid abdomen may associate with diaphragmatic hernia.

B.Auscultation: Listen for bowel sound

C.Palpation: Palpate for liver and spleen

XV. Umbilicus: Normally the umbilicus has 2 arteries and 1 vein.

XVI. Genitalia: Any infant with ambiguous genitalia should not undergo gender assignment till a formal endocrinological evaluation.

XVII. Anus and rectum: Check for patency of the anus. Meconium should pass within 48 hours

XVIII. Extremities: Syndactyly, Polydactyly, Simian crease should be noted.

A.Pes Varus: most of this condition is due to intrauterine compression. It will resolve spontaneously months later.

B.Pes vulgas

C.Talipes equinovarus (club foot) is common in males.

XIX. Trunk and spine: Check for any gross defects of the spine.

XX. Hips: Evaluate for congenital hip dislocation

XXI. Nervous system:

First, observe the infant for any abnormal movement or excessive irritability.

A. Muscle tone: Hypotonia, Hypertonia

B. Reflexes: Rooting reflex, Blink reflex, Grasp reflex, Tonic neck reflex, Moro reflex

C. Movement: Check for spontaneous movement of limbs, trunk, face, and neck. A fine tremor is usually normal. Clonic movements are not normal and may be seen in seizures.

D. Peripheral nerves:

1. **Erb-Duchene paralysis**
2. **Klumpke's paralysis**

XIV-7. 預防接種時程表

適合接種年齡	疫苗種類	
出生 24 小時內盡速接種	B 型肝炎免疫球蛋白(HBIG) (若母親為 HBsAg(+)才需施打)	一劑
	B 型肝炎疫苗(HepB 1st)	第一劑
出生滿 1 個月	B 型肝炎疫苗(HepB 2nd)	第二劑
出生滿 2 個月	五合一疫苗(DTap-Hib-IPV 1st) (白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺混合疫苗)	第一劑
	13 價結合型肺炎鏈球菌疫苗 (PCV 13 1st)	第一劑
出生滿 4 個月	五合一疫苗(DTap-Hib-IPV 2nd) (白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺混合疫苗)	第二劑
	13 價結合型肺炎鏈球菌疫苗 (PCV 13 2nd)	第二劑 (與前劑至少間隔 8 週)
出生滿 5 個月 (5-8 個月)	卡介苗(BCG)	第一劑
出生滿 6 個月	B 型肝炎疫苗(HepB 3rd)	第三劑
	五合一疫苗(DTap-Hib-IPV 3rd) (白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺混合疫苗)	第三劑
出生滿 12 個月	麻疹腮腺炎國麻疹混合疫苗(MMR 1st)	第一劑
	水痘疫苗(Varicella)	一劑
出生滿 12-15 個月	13 價結合型肺炎鏈球菌疫苗 (PCV 13 3rd)	第三劑
	A 型肝炎疫苗(HepA 1st)	第一劑
出生滿 15 個月	日本腦炎疫苗(JE 1st)(活性減毒疫苗)	第一劑
出生滿 18 個月	五合一疫苗(DTap-Hib-IPV 4th) (白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺混合疫苗)	第四劑
出生滿 18-21 個月	A 型肝炎疫苗(HepA 2nd)	第二劑(與前劑至少間隔 6 個月)
出生滿 27 個月	日本腦炎疫苗(JE 2nd) (活性減毒疫苗)	第二劑(與前劑至少間隔 12 個月)
滿 5 歲至入國小前	四合一疫苗(DTap-IPV) (白喉破傷風非細胞性百日咳及不活化小兒麻痺混合疫苗	一劑
	麻疹腮腺炎國麻疹混合疫苗(MMR 2nd)	第二劑
	(日本腦炎疫苗(活性減毒疫苗))*針對完成 3 劑舊劑型的不活化疫苗之幼童	一劑(與前劑至少間隔 12 個月)

*針對完成 3 劑舊劑型的不活化疫苗之幼童，於滿 5 歲至入國小前再接種 1 劑，與前一劑疫苗間隔至少 12 個月。

※自費疫苗: 13 價肺炎鏈球菌疫苗(可於 6 個月大時自費補打一隻)、口服輪狀病毒疫苗(出生滿 6 週至 8 個月)

※其他疫苗: 流感疫苗(出生滿 6 個月即可施打，每年集中於 10 月至 12 月接種。8 歲（含）以下兒童，若是初次接種，應接種 2 劑，2 劑間隔 4 週以上。9 歲以上則不論過去季節性流感疫苗接種史，都只須接種 1 劑。

Reference: 衛生福利部疾病管制署 專業版 , 107. 08 版 (2018/9/10 查詢)

我國現行兒童預防接種時程

107.08 版

接種年齡 / 疫苗	24hr 內儘速	1 month	2 months	4 months	5 months	6 months	12 months	15 months	18 months	21 months	24 months	27 months	滿 5 歲至入國小前	國小學童
B 型肝炎疫苗(HepB)	**第一劑**	**第二劑**				**第三劑**[5]								
卡介苗(BCG)[1]					**一劑**									
白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺五合一疫苗(DTaP-Hib-IPV)			**第一劑**	**第二劑**		**第三劑**[5]			**第四劑**					
結合型肺炎鏈球菌疫苗(PCV13)			**第一劑**	**第二劑**			**第三劑**							
水痘疫苗(Varicella)							**一劑**							
麻疹腮腺炎德國麻疹混合疫苗(MMR)							**第一劑**						**第二劑**	
日本腦炎疫苗(JE)[2]								**第一劑**				**第二劑**	**一劑***	
流感疫苗(Influenza)[3]						←			**初次接種二劑，之後每年一劑**					→
A 型肝炎疫苗(HepA)[4]							**第一劑**		**第二劑**					
白喉破傷風非細胞性百日咳及不活化小兒麻痺混合疫苗(DTaP-IPV)													**一劑**	

1.105年起，卡介苗接種時程由出生滿24小時後，調整為出生滿5個月(建議接種時間為出生滿5-8個月)。

2.106年5月22日起，改採用細胞培養之日本腦炎活性減毒疫苗，接種時程為出生滿15個月接種第1劑，間隔12個月接種第2劑。

*針對完成3劑不活化疫苗之幼童，於滿5歲至入國小前再接種1劑，與前一劑疫苗間隔至少12個月。

3.8歲(含)以下兒童，初次接種流感疫苗應接種2劑，2劑間隔4週。

4.A型肝炎疫苗107年1月起之實施對象為民國106年1月1日（含）以後出生，年滿12個月以上之幼兒。另包括設籍於30個山地鄉、9個鄰近山地鄉之平地鄉鎮及金門連江兩縣等原公費A肝疫苗實施地區補接種之學齡前幼兒。

5.106年5月1日起，以六合一疫苗暫用以取代嬰幼兒應接種之第3劑B型肝炎疫苗及五合一疫苗。

各項常規疫苗最小接種年齡與最短接種間隔

107.02 版

疫　　苗	建議接種時程	第 1 劑最小接種年齡[1]	最短接種間隔[1]		
			第 1 與第 2 劑	第 2 與第 3 劑	第 3 與第 4 劑
卡介苗(BCG)	5-8 個月	出生後			
B 型肝炎疫苗(HepB)	出生 24 小時內儘速 1 個月 6 個月#	出生 24 小時內儘速	4 週	8 週[2] (且第 1、第 3 劑應間隔至少 16 週)	
白喉破傷風非細胞性百日咳、b 型嗜血桿菌及不活化小兒麻痺五合一疫苗(DTaP-Hib-IPV)	2 個月 4 個月 6 個月# 18 個月	6 週	4 週	4 週	6 個月[3] (滿 1 歲後)
13 價結合型肺炎鏈球菌疫苗(PCV13)[4]	2 個月 4 個月 12-15 個月	6 週	8 週	8 週 (滿 1 歲後)	
水痘疫苗(Varicella)[5]	12 個月	12 個月			
麻疹腮腺炎德國麻疹疫苗(MMR)[5]	12 個月 滿 5 歲	12 個月[6]	4 週		
日本腦炎活性減毒疫苗(JE)[5]	15 個月 27 個月	12 個月	12 個月		
日本腦炎不活化疫苗(JE)	15 個月 16 個月 28 個月 滿 5 歲	12 個月	4週	6 個月	6 個月 (滿 4 歲後)
破傷風白喉非細胞性百日咳及不活化小兒麻痺疫苗(DTaP-IPV /Tdap-IPV)[7]	滿 5 歲	4 歲			
A 型肝炎疫苗(HepA)[8]	12-15 個月 18-21 個月	12 個月	6 個月		

#106年5月1日起，以六合一疫苗暫用以取代嬰幼兒應接種之第3劑B型肝炎疫苗及五合一疫苗。

附註：

1. 接種間隔標示"月"係日曆月，4週=28天。
2. **B型肝炎疫苗第2劑與第3劑之最短接種間隔為8週，且第3劑與第1劑應間隔至少16週**。
3. 五合一疫苗第4劑如於出生滿12個月時即接種，不適用再提前4天之寬限。
4. **PCV13第1劑與第2劑間隔至少8週，第3劑應在滿1歲後接種，且與第2劑間隔至少8週**；PCV13第1劑如於出生滿7個月後才開始接種，與第2劑之接種間隔可縮短至4週。
5. **水痘、MMR與日本腦炎活性減毒等疫苗可同時接種分開不同部位，若未同時接種應至少間隔28天，不能再縮短。**
6. 如為因應疫情或須前往麻疹流行地區，MMR第1劑最早可提前於出生滿6個月後接種，但**滿12個月後仍須按常規時程完成2劑公費MMR疫苗**。
7. 五合一疫苗第4劑若於4歲以後才接種，滿5歲之DTaP-IPV /Tdap-IPV疫苗可不再接種。
8. **A型肝炎疫苗107年1月起之實施對象為民國106年1月1日（含）以後出生，年滿12個月以上之幼兒。另包括設籍於30個山地鄉、9個鄰近山地鄉之平地鄉鎮及金門連江兩縣等原公費A肝疫苗實施地區補接種之學齡前幼兒。**
9. 表列**同種類疫苗(水痘、MMR及JE活性減毒疫苗除外)各劑次間**，**因特殊情況**可容許最短接種間隔或最小接種年齡再提前4天之寬限期，若提前5天(含)以上接種者，則該劑疫苗應視同無效，並依表列接種年齡、間隔規範重新安排接種。

XV. 體表面積

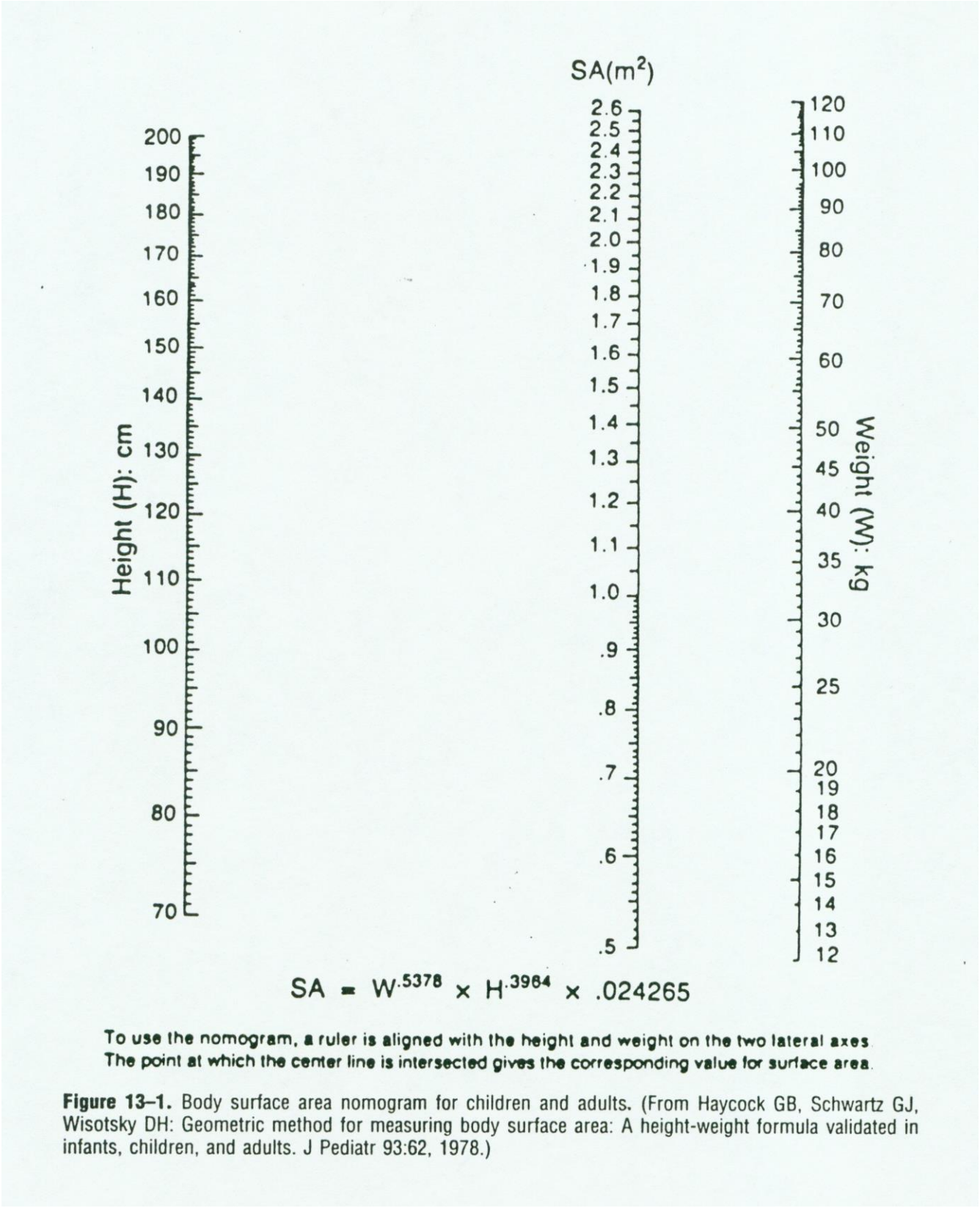


Figure 13–1. Body surface area nomogram for children and adults. (From Haycock GB, Schwartz GJ, Wisotsky DH: Geometric method for measuring body surface area: A height-weight formula validated in infants, children, and adults. J Pediatr 93:62, 1978.)

XVI-1. 敗血症的定義及抗生素的選用

Definitions of the Systemic Inflammatory Response Syndrome

Bacteremia:

The presence of viable bacteria in the blood. The presence of other organisms(fungi, viruses) should be modified to reflect those conditions(ie,fungemia, viremia).

Systemic inflammation response syndrome(SIRS):

A clinical syndrome that represents the body's response to a wide variety of severe insults. The condition is manifested by the presence of at least two of the following conditions:

1. Temperature>38°C or <36°C(>100.4°F or <96.8°F)
2. Tachycardia(heart rate > 160 beats/min for infants, >150 beats/min for children)
3. Tachypnea(respiratory rate >60 breaths/min for infants, >50 breaths/min for children) or $PaCO_2$ <32 torr.
4. White blood cell count > 12,000 cells/mm3 or >10% immature(band) forms.

Sepsis: The systemic response to infection manifested by two or more of the criteria for SIRS.

Severe sepsis:

Sepsis associated with hypotension (systolic blood pressure <65 mm Hg in infants, <75 mm Hg in children or <5th percentile for age, <90 mm Hg in adolescents, or a reduction of >40 mm Hg from the baseline in the absence of other causes for hypotension), hypoperfusion, or organ dysfunction. Hypoperfusion and perfusion abnormalities may include, but are not limited to, lactic acidosis, oliguria, hypoxemia, or acute alteration in mental status (Glasgow Coma Scale score 3 below baseline).

Septic shock:

Sepsis with hypotension, despite adequate fluid resuscitation, with the presence of perfusion abnormalities that may include, but are not limited to, lactic acidosis, oliguria, or acute alteration in mental status.

Note: Patients who are on inotropic or vasopressor agents may not be hypotensive at the time perfusion abnormalities are measured.

Multiple organ dysfunction syndrome:

Presence of altered organ function in an acutely in patient such that homeostasis cannot be maintained without intervention.

From American College of Chest Physicians/Society of Critical Cure Medicine Consensus Conference. Pediatric modification of vital signs are based on Jufari and McCracken. Pediatric Infect Dis J. 1992;11:739-749.

XVI-2.
Antimicrobial Choices for Empiric Therapy in Septic Shock及常用抗生素劑量

Neonate	**Ampicillin plus aminoglycoside** or cefotaxime; if nosocomial, add vancomycin
Child	**Cefotaxime or ceftriaxone plus vancomycin**; * if nosocomial, vancomycin plus an antibiotic against resistant Gram-negative bacteria specific to the institution such as: •Antipseudomonal cephalosporin (ceftazidime or cefepime) •Aminoglycoside (gentamicin, tobramycin) •Extended-generation penicillin with beta-lactamase inhibitor (pipracillin/sulbactam) •Carbapenem (imipenem or meropenem)
Invasive group A streptococci(consider following varicella infection)	**Penicillin AND clindamycin**
Herpes or varicella	**Acyclovir**
**In cases of suspected Gram-positive infection in an area that has increased staphylococcal or pneumococcal resistance or for patients who have been treated with antibiotics frequently (sickle cell anemia, frequent otitis media).*	

TABLE 16: PEDIATRIC DOSAGES OF SELECTED ANTIBACTERIAL AGENTS

[Adapted from: (1) Nelson's Pocket Book of Pediatric Antimicrobial Therapy, 2002-2003, 15th Ed., J. Bradley & J. Nelson, eds., Lippincott Williams and Wilkins, and (2) 2003 Red Book, 26th Ed., American Academy of Pediatrics, pages 700–718

DRUG	DOSES IN MG/KG/D OR MG/KG AT FREQUENCY INDICATED[1]				
	BODY WEIGHT <2000 gm		BODY WEIGHT >2000 gm		>28 DAYS OLD
	0–7 days old	8–28 days old	0–7 days old	8–28 days old	
Aminoglycosides, IV or IM (check levels; some dose by gestational age + wks of life; *see Nelson's Pocket Book, page 19*)					
Amikacin	7.5 q18–24h	7.5 q12h	10 q12h	10 q12h	10 q8h
Gent/tobra	2.5 q18–24h	2.5 q12h	2.5 q12h	2.5 q12h	2.5 q8h
Aztreonam, IV	30 q12h	30 q8h	30 q8h	30 q6h	30 q6h
Cephalosporins					
Cefaclor					20–40 div tid
Cefadroxil					30 div bid (max 2 g/d)
Cefazolin	20 q12h	20 q12h	20 q12h	20 q8h	20 q8h
Cefdinir					7 q12h or 14 qd
Cefepime					150 div q8h
Cefixime					8 as qd or div bid
Cefotaxime	50 q12h	50 q8h	50 q12h	50 q8h	50 q6h (75 q6h for meningitis)
Cefoxitin			20 q12h		80–160 div q6h
Cefpodoxime					10 div bid (max 400 mg/d)
Cefprozil					15–30 div bid (max 1 g/d)
Ceftazidime	50 q12h	50 q8h	50 q12h	50 q8h	50 q8h
Ceftibuten					4.5 bid
Ceftizoxime					33–66 q8h
Ceftriaxone	50 qd	50 qd	50 qd	75 qd	50–75 qd (meningitis 100)
Cefuroxime IV	50 q12h	50 q8h	50 q8h	50 q8h	50 q8h (80 q8h for meningitis)
po					10–15 bid (max 1 g/d)
Cephalexin					25–50 div 4x/d (max 4 g/d)
Loracarbef					15–30 div bid (max 0.8 g/d)
Chloramphen. IV	25 q24h	25 q24h	25 q24h	15 q12h	12.5–25 q6h (max 2–4 g/d)
Clindamycin IV	5 q12h	5 q8h	5 q8h	5 q6h	7.5 q6h
po					5–6 q8h
Ciprofloxacin po[2]					20–30 div bid (max 1.5 g/d)
Imipenem[3] IV			25 q12h	25 q8h	15–25 q6h (max 2 g/d)
Linezolid	No data	10 q8h	No data	10 q8h	10 q8h to age 12
Macrolides					
Erythro IV & po	10 q12h	10 q8h	10 q12h	13 q8h	10 q6h
Azithro po					10–12 day 1, then 5/d[4]
Clarithro po					7.5 q12h (max. 1 g/d)
Meropenem IV	20 q12h	20 q8h	20 q12h	20 q8h	60–120 div q8h (120 for meningitis)
Metro IV & po	7.5 q24h	7.5 q12h	7.5 q12h	15 q12h	7.5 q6h
Penicillins					
Ampicillin	50 q12h	50 q8h	50 q8h	50 q6h	50 q6h
Amp-sulbactam					100–300 div q6h
Amoxicillin po				30 div bid	25–50 div tid
Amox-Clav po	30 div bid	30 div bid	30 div bid	30 div bid	45 or 90 (AM/CL-HD) div bid if over 12 wks of age
Cloxacillin					50–100 div 4x/d
Dicloxacillin					12–25 div 4x/d
Mezlocillin	75 q12h	75 q8h	75 q12h	75 q8h	75 q6h
Nafcillin,oxacillin IV	25 q12h	25 q8h	25 q8h	37 q6h	37 q6h (to max. 8–12 gm/d)
Piperacillin, PIP/tazo IV	75 mg/kg q12h	75 mg/kg q12h	75 mg/kg q8h	75 mg/kg q8h	100–300 div q4–6h
Ticarcillin, T.clav IV	75 q12h	75 q8h	75 q8h	75 q6h	75 q6h
Penicillin G, **U/kg** IV	50,000 q12h	75,000 q8h	50,000 q8h	50,000 q6h	50,000 U/kg/d
Penicillin V					25–50 mg/kg/d div 3–4x/d
Rifampin po			10, single dose	20, single dose	20, single dose (max. 600 mg)
Sulfisoxazole po				120–150	120–150 mg/kg/d div q4–6h
TMP/SMX po,IV; UTI: 8–12 TMP component div bid; Pneumocystis: 20 TMP component div 4x/d					
Tetracycline po (age 8 or older)					25–50 div 4x/d
Doxycycline po,IV (age 8 or older)					2–4 div bid
Vancomycin IV	12.5 q12h	15 q12h	18 q12h	22 q12h	40–60 div q6h

Abbreviations: **Chloramphen** = chloramphenicol; **Clav** = clavulanate; **div** = divided; **Gent/tobra** = gentamicin/tobramycin; **Metro** = metronidazole; **Tazo** = tazobactam; **TMP** = trimethoprim; **TMP/SMX** = trimethoprim/sulfamethoxazole; **UTI** = urinary tract infection

[1] May need higher doses in patients with meningitis: *see CID 39:1267, 2004*
[2] With exception of cystic fibrosis, not approved for use under age 18.
[3] Not recommended in children with CNS infections due to risk of seizures.
[4] Dose for otitis; for pharyngitis, 12 mg/kg x5 d.

XVI-3.

Suggested Duration of Antibiotic Therapy for Specific Pediatric Infections

Infection	Duration (days)	(Range)
Neonatal		
• Sepsis	10-14	(7-≥14)
• Meningitis		
• Group B streptococcus	14-21	(≥10-≥21)
• Gram-negative	21	(14-≥21)
Beyond three months of age		
• Meningitis		
• N. menigitidis	7	(5-14)
• S. pneumoniae	10	(7-14)
• H. influenzae	10	(7-14)
• Gram-negative	21	(14-28)
• Group B streptococcus	14	(7-21)
• Pneumonia		
• S. pneumoniae	7-10	(≥5-14)
• H. influenzae	10-14	(7-≥21)
• S. aureus	≥21	
• C. trachomatis	14-21	(10-21)
• Empyema	≥21	(≥14-28)
• Acute otitis media	10	(5-14)
• Acute sinusitis	10	(7-21)
• Septic arthritis (S. aureus)	≥28	(14-≥28)
• Osteomyelitis (S. aureus)	≥28	(≥21-≥28)
• Acute pyelonephritis	10-14	

Bold indicates most commonly cited duration，台兒醫誌第 41 卷增刊 2000，卓正宗教授

XVII-1. 生命徵象(Vital Signs)正常值 (1)

								Blood Pressure		
								Systolic/	Mean	Systolic
	Respiratory Rate			Heart Rate				diastolic	(ave)	diastolic
	Normal	Bradypnea	Tachypnea	Normal	Bradycadia	Tachycardia		(ave)		(range)
Newborn	30-50	<30	>50	125	<70	>190	Newborn			
(<1 wk)							>600 gm	45/20	25	35-55
and										14/35
Newborn							>1000gm	48/25	35	36-58
(1wk-1mo)										17-38
							>2000gm	50/30	40	42-66
										22-42
							>3000gm	50/35	45	42-66
										25-48
							>4000gm	65/40	50	55-80
										30-50
							Newborn	75/50	60	60-90
							(>12hr)			20-60

XVII-2. 生命徵象(Vital Signs) 正常值(2)

									Blood Pressure		
									Systolic/	Mean	Systolic/
	Respiratory Rate			Heart Rate					Diastolic	(ave)	Diastolic
	Normal	Bradypnea	Tachypnea	Normal	Bradycadia	Tachycardia			(ave)		(range)
Infant (1 mo-1 1/2 yr)	25-35	<25	>35	120	<80	>160			90/60	70	74-100 50-70
Child (1 1/2-5 yr)	20-30	<20	>30	100-120	<80	>120-160			100/60	75	82-110
											50-78
(5 yr-12 yr)	15-25	<15	>25	90-100	<70	>115			110/60	80	84-120
											54-80
Adolascent)	12-20	<12	>20						120/70	85	94-140
(>12yr)											62-88
			Girl	Boy	Girl	Boy	Girl	Boy			
		12yr	90	85	70	65	110	105			
		14yr	85	80	65	60	105	100			
		16yr	80	75	60	55	100	95			
		18yr	75	70	55	50	95	90			
			Normal		**Bradycardia**		**Tachycardia**				

XVIII-1. 生命徵象(Vital signs)簡易對照表

正常呼吸速率及心跳

1. Hypotension is age dependent: 連續 BP 的變化也重要

   systolic BP = < 70 + (2 × age in years)

   ↓ perfusion = capillary refill > 2 sec.

2. Average Heart Rates at Rest (beats per min)

Age	Average	Bradycardia	Tachycardia
NB	125	70	190
1-11mo	120	80	160
2 yr	110	80	130
4 yr	100	80	120
6 yr	100	75	115
8 yr	90	70	110
10 yr	90	70	110

	F	M	F	M	F	M
12 yr	90	85	70	65	110	105
14 yr	85	80	65	60	105	100
16 yr	80	75	60	55	110	95
18 yr	75	70	55	50	95	90

3.

Respiratory Rates

Age	Norm Rate/Min	Bradypnea	Tachypnea
NB	30-50	< 30	> 50
1 yr	25-35	< 25	> 35
2-4 yrs	20-30	< 20	> 30
5-15 yrs	15-25	< 15	> 25
15-20 yrs	12-20	< 12	> 20

* count for a full 60 sec to detect irrgularities

* respiratory rates may be > 50 for first hour to two after birth

XVIII-2. 急救藥物

CARDIOPULMONARY RESUSCITATION			Age	Newborn	Infant	1 yr	3 yr	6 yr	10 yr	Adol	Adult
			Wt	**3 kg**	**5 kg**	**10 kg**	**15 kg**	**20 kg**	**30 kg**	**50 kg**	**70 k**
			ETT+ Size++	3.0-3.5	3.5-4.0	4.0-4.5	4.5-5.0	5.0-5.5	6.0-6.5cuffed	7.0-7.5cuffed	7.5-8.0cuffed
Drug/indication	Route	Dose(conc*)	Freq	**Dose (based on above age/weight)**							
EPINEPHRINE •Bradycardia •Asystole/pulseless arrest(first dose)	IV IM IO ET: see high-dose epinephrine below	0.01 mg/kg (1:10,000=0.1 mg/ml) 0.1 ml/kg	Q3-5 min	0.3 ml	0.5 ml	1 ml	1.5 ml	2 ml	3 ml	5 ml	10 ml
HIGH-DOSE EPINEPHRINE •All ETT does	IV IO ET	0.1 mg/kg **(1:1000=0.1 mg/ml)** 0.1 ml/kg	Q3-5 min	0.3 ml	0.5 ml	1 ml	1.5 ml	2 ml	3 ml	5 ml	10 ml
ATROPINE SULFATE •Bradycardia •AV block	IV IM ET IO	0.02 mg/kg (0.1mg/ml) 0.2 ml/kg	Q5 min	1 ml minimum dose	1 ml	2 ml	3 ml	4 ml	5 ml (max dose child)	10 ml (max dose adol)	10 ml (max dose 2mg)
BICARBONATE •Metabolic acidosis w/good ventilation •Hyperkalemia	IV IO	1 mEq/kg **Newborn: 0.5mEq/ml, 2ml/kg** Adult: 1 mEq/ml, 1 ml/kg	Q10 min	3 mEq	5 mEq	10 mEq	15 mEq	20 mEq	30 mEq	50 mEq	70 mEq

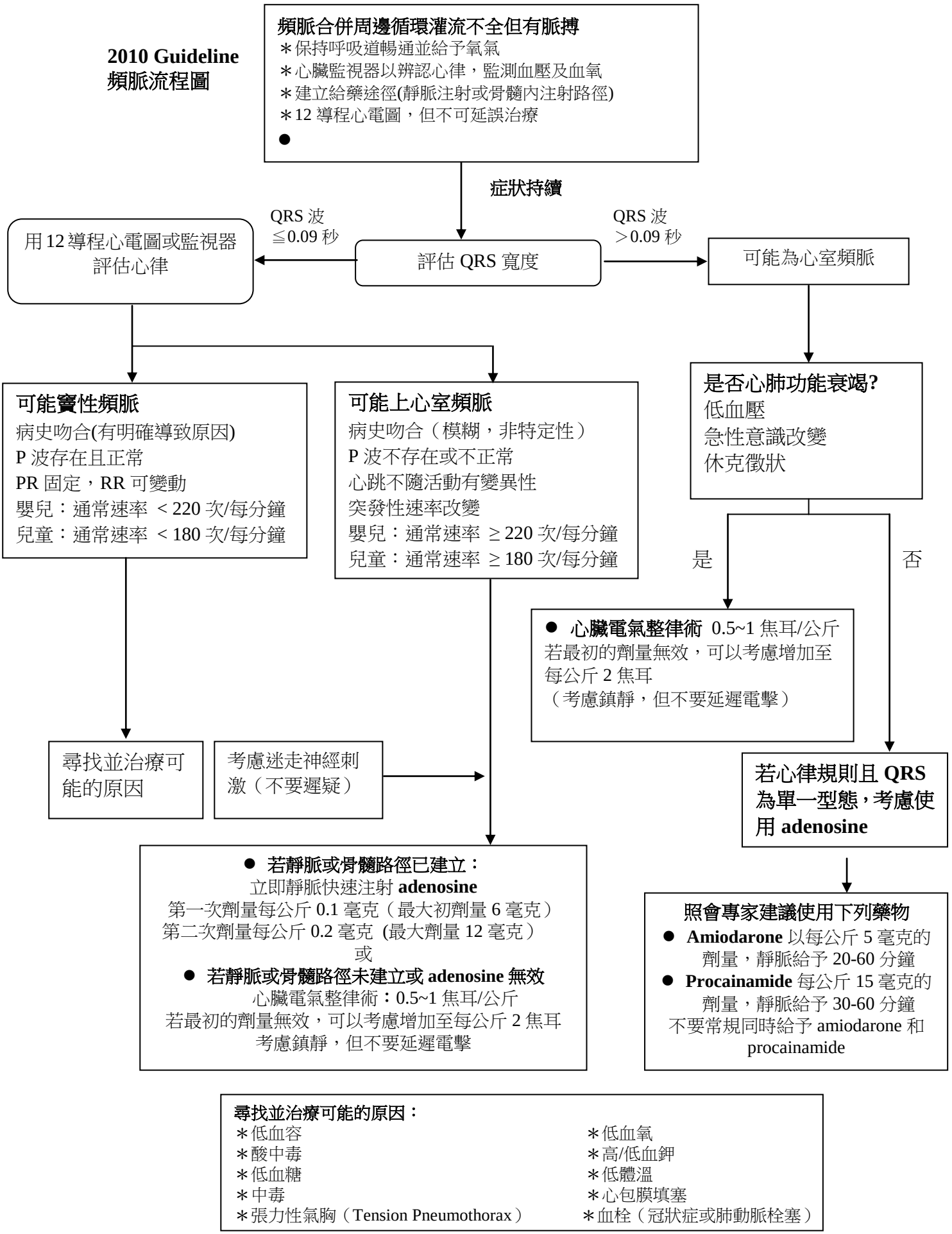
NARCAN: 0.1 mg/kg/dose IM/IV/SC to **max** of 2 mg/dose. May repeat Q2-3 min.
OXYGEN: 100%
GLUCOSE: If hypoglycemic: 2-4 ml/kg bolus of $D_{25}W$, then constant infusion of glucose at 8 mg/kg/min.
CALCIUM: If hypocalcemic, hyperkalemic, or Ca channel blocker toxic: 100 mg/kg/ 10% Ca gluconate, or 20 mg/kg 10% Ca chloride.
DEFIBRILLATION: 2 Joules/kg, may double and repeat.
CARDIOVERSION: Initial: 0.5 Joules/kg. May double and repeat.
ADENOSINE: 0.1 mg/kg rapid IV push. Max total dose=12 mg.
*Beware: Standard drug concentrations may vary.
+ETT size(mm)=[16+age(yr)] / 4
++internal diameter +approx. distance of insertion = i.d. × 3
"*LANE*" drugs may be given by ETT: Lidocaine,
Atropine, Naloxone, Epinephrine. Dilute medication with normal saline to volume of 3-5 ml. Follow with positive pressure ventilations

IV INFUSIONS:

6×{desired dose(mcg/kg/min)/desired rate(ml/hr)}×wt(kg)=mg drug/100 ml fluid

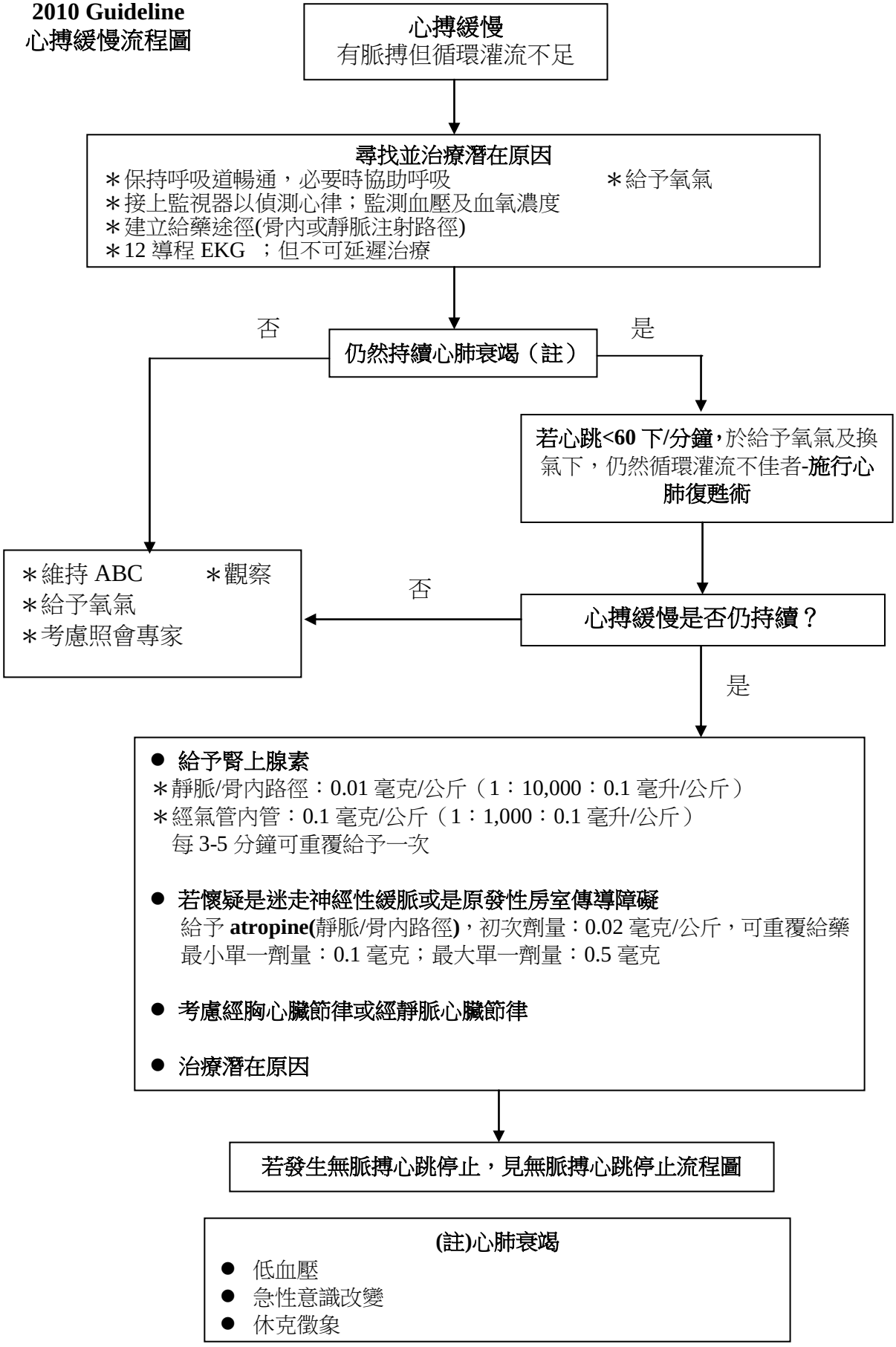
Medication	Usual Dose, mcg/kg/min	Dilution in 100 ml D_5W	IV Infusion Rate
Dopamine	2-20	6 mg/kg	1 ml/hr=1 mcg/kg/min
Dobutamine	2-20	6 mg/kg	1 ml/hr=1 mcg/kg/min
Epinephrine	0.1-0.4	0.6 mg/kg	1 ml/hr=0.1 mcg/kg/min
Isoproterenol	0.1-2	0.6 mg/kg	1 ml/hr=0.1 mcg/kg/min
Lidocaine	20-50	6 mg/kg	1 ml/hr=1 mcg/kg/min

2010 Guideline
頻脈流程圖



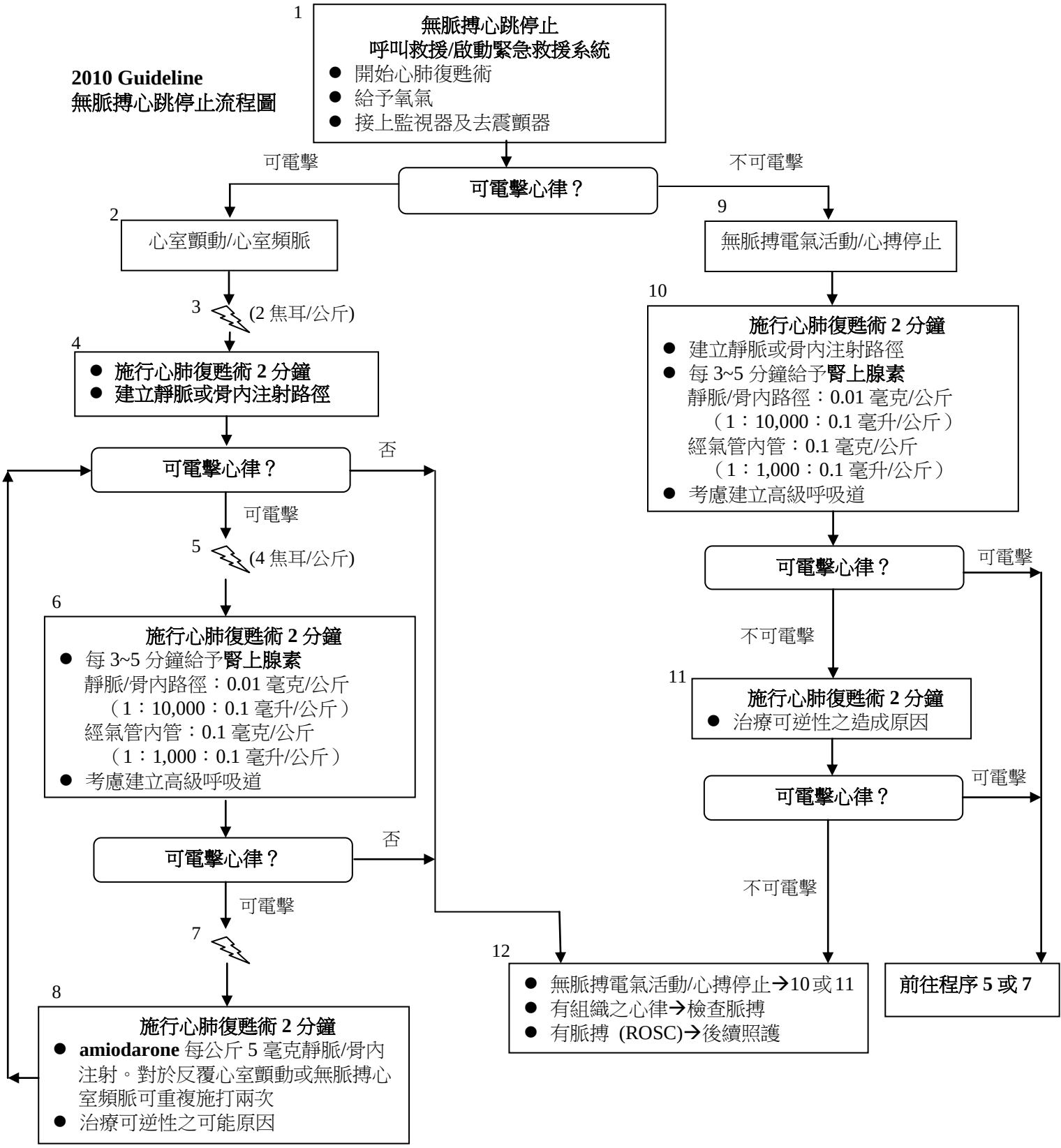
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2010 Guideline
心搏緩慢流程圖



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2010 Guideline
無脈搏心跳停止流程圖



急救過程中

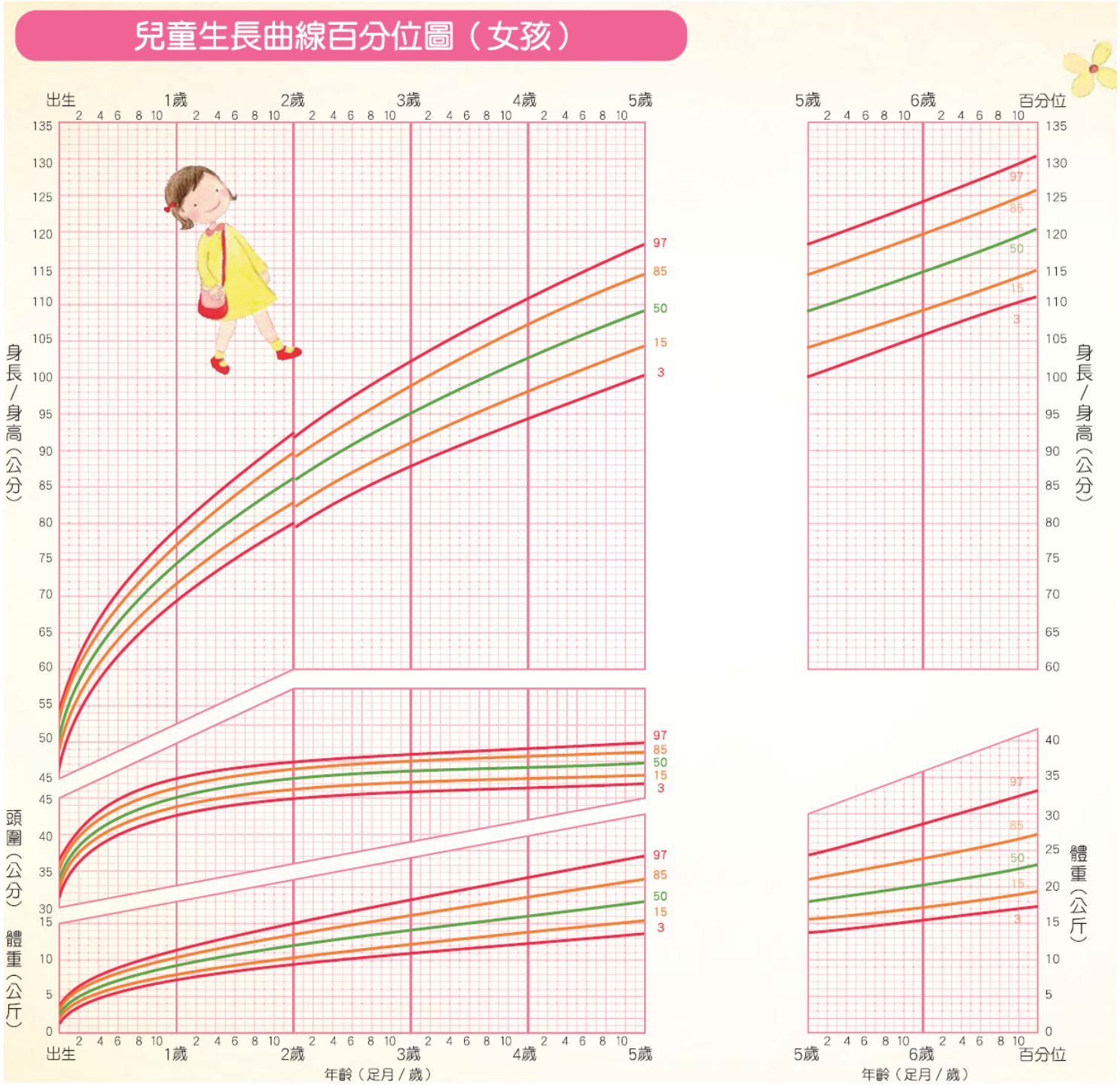
- 在心肺復甦過程中，快點壓（每分鐘 100 次以上），用力壓 (≧1/3 胸前後徑)
- 讓胸廓有足夠的時間回彈
- 減少中斷按壓的時間
- 1 個週期心肺復甦術：按壓 15 下後，吹 2 口氣；5 個週期約略 2 分鐘
- 避免過度換氣
- 若無高級呼吸道，按壓吹氣比為 15:2。若有高級呼吸道，壓胸不間斷，每分鐘吹 8~10 口氣。
- 每兩分鐘替換施行胸部按壓的施救者
- 以 ETCO2 確認氣管內管位於氣管中。

可能原因包括

＊低血容	＊低血氧
＊酸中毒	＊高/低血鉀
＊低血糖	＊低體溫
＊中毒	＊心包膜填塞
＊張力性氣胸（Tension Pneumothorax）	
＊血栓（冠心症或肺動脈栓塞）	

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兒童生長曲線百分位圖（女孩）



兒童生長曲線百分位圖（男孩）





出生 1歲 2歲 3歲 4歲 5歲

身長／身高（公分）

頭圍（公分） 體重（公斤）

97 85 50 15 3

年齡（足月／歲）

5歲 6歲 百分位

身長／身高（公分）

97 85 50 15 3

5歲 6歲 百分位

頭圍（公分） 體重（公斤）

97 85 50 15 3

年齡（足月／歲）

IXX. 簡易兒童發展量表—長庚醫院復健科林源圖及王錦滿醫師發展

1. 適用 4 個月-2 歲
2. 採簡單圖畫及文字(圖略)
3. 分成四大領域，共 10 分項，配合疫苗注射時間，分為 4、6、9、12、18 及 24 個月六大追蹤階段
4. 打分數：出現行為給予 10 分，共 10 分項，故滿分 100 分，有下列情形請轉介給專科醫師，以早期診斷發展遲緩
   ①得分在 80 分以下(有二項不通過)
   ②有任何 2 個測試不通過落在同一分項中，且持續到下一個追蹤年齡

	4 個月	6 個月	9 個月	1 歲	18 個月	24 個月
粗動作	1.俯臥時骨盆平貼於床面，頭、胸部可抬離床面 2.拉扶坐起，只有輕微頭部落後 3.坐姿扶持，頭部幾乎一直抬起	1.抱直時，脖子豎直頸保持在中央 2.會自己翻身(俯臥→仰臥) 3.可以自己坐在有靠背椅子上	1.不須扶持可坐穩 2.獨立自己爬 3.坐時，會移動身體挪向所要的物體	1.雙手扶著傢俱會走幾步 2.拉著雙手會移幾步 3.拉著物體自己站起來	1.可以走很快 2.走的很穩 3.牽著他或扶欄干可上樓梯	1.自己會上下樓梯 2.自己會由椅子上爬下 3.會踢球(一腳站立)
細動作	1.手會自動張開 2.常舉手作”熟視手部” 3.當搖鈴被放到手中會握住 1 分鐘	1.雙手互握一起 2.手能伸向物體 3.自己會拉開在他臉上的手帕	1.將東西由一手換到另一手 2.用兩手拿小杯子 3.自己會抓住東西往嘴裡送	1.拍手 2.會把一些小東西放入杯子 3.會撕紙	1.用筆亂塗 2.打關瓶蓋 3.開始較常用特別一邊的手	1.重疊兩塊積木 2.一頁一頁翻圖畫表 3.將杯子水倒至另一個杯子
語言溝通	1.轉頭偏向音源 2.有人同他說話會咿呀作聲	1.哭鬧時，會因媽媽的安撫聲而停止哭 2.看他時，會回看你的眼睛	1.轉向聲源 2.會發生單音如”ㄇㄚ“ “ㄅㄚ”	1.以揮手表示再見 2.會摹仿簡單聲音	1.有意義的叫爸媽 2.會跟著或主動說出一個單字	1.指出身體之一部分 2.至少講 10 個單字
身邊處理及社會性	1.雙眼可凝視人物並追尋移動之物 2.會對媽媽親切露出微笑	1.逗他會微笑 2.餵他吃時，會張口或用其他動作表示要吃	1.自己拿餅乾吃 2.會怕陌生人	1.叫他，他會來 2.會脫帽子	1.雙手端杯子喝水 2.替他穿衣服時會自動伸出胳臂或腿	1.自己脫衣服 2.會打開糖果紙

XX. 臨床上實用的一些計算工具程式介紹

自己在臨床的工作上常常遇到一些 routine 要計算或查表的情形，比如說病人一入院就要做的生長評估、NICU 一些繁雜的計算、繪製家族史、將 X 光片印到病例上、整理病人 LAB DATA 的變化、DKA 病人的處理、脫水合併低血鈉、高血鈉的處理...等，常常需要花不少的時間，但是其實這些都可以交給電腦幫我們計算，我們只要幫電腦驗算一下就可以使用了。因此斷斷續續寫了以下幾個工具，分享給大家使用，希望對臨床的工作能有所助益。這些工具已經請秘書放在兒童內科部首頁的最新通告裡了。以下做個簡介，以後會陸續將詳細的說明到網路上。

一、生長評估工具，新生兒評估工具...Date v.2.18

輸入生日、歲數、週數或預產期，選擇性別，輸入身高體重及頭圍，按下生長評估（或新生兒評估）程式便會評估其身高體重頭圍的 percentile。目前版本另外可以繪製單一或序列的生長曲線，直接列印後為病歷紙的大小，可直接夾病歷。

日期輸入的方法有很多種，例如西元 2006 年 8 月 20 日可以輸入成：2006/8/20 或 0950820 或 950820 或 06-08-20 或 August 20, 2006 或 Aug 20,2006。三歲五個月大可以輸入 3y5m，38 週以輸入 38w，二十五天可以輸入 25d 依此類推。

二、NICU 計算工具...NICU v.2.32

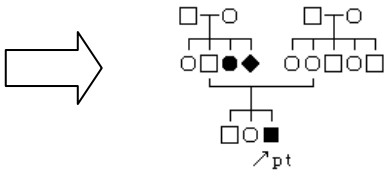
有 NICU 所需要的大部分計算功能，並有智慧型輸入功能「★★」。功能較為複雜，詳情請見說明書。想要使用本計算工具的同仁，務必先詳閱此說明書，謝謝。

三、家族史產生器...Pedigree v.1.50

用來繪製三代的家族史文字圖，詳細說明請見程式本身，下面是一個範例：

代碼：

```
OFQV=OMZOZ
ZOB(E)
```



四、放置圖片的工具...PIC v.4.21

選擇圖檔，輸入印出來時的寬（mm）高（mm）距頁面左邊界距離，距上邊界距離

按下「加圖」「預覽」「放圖至新網頁」然後在產生的新網頁中選檔案→設定列印格式→列印即可。

五、LAB DATA 整理工具...LAB v.2.12

利用檢驗查詢中的「複製報告」→「門住診（粗略）」→「該檢驗組別全部報告」得到的文字資料貼入程式中，按下「序列整理(組別粗略全複製之資料)」便可以將檢驗資料整理成表格，

將院內 Lab Data 整理成表格的工具，要將表格存回家存檔時請選：「網頁，僅 HTML」的存檔類型即可，該檔案可以被 excel，word，power point 等 OFFICE 程式所相容讀取。

六、DKA 在 ICU 的醫囑程式...Two Bag System Order Form v.1.03

這是 DKA 在 ICU 的醫囑程式，本醫囑程式，已經經過 PICU 學長測試驗算過，已實際應用於臨床工作上，大致上沒有問題，但是實際應用上還是請大家要稍微小心一下。詳細說明已在程式本身中了。

七、輸液治療計算工具...Pediatric Fluids and Electrolyte Therapy v1.01

輸入目前的體重、生病前的體重或 dehydration 的程度，生病時間的長短，Na 的檢驗值，程式會算出一個輸液治療的參考，千萬要視情形調整，以免造成 over hydration。

Memo

Memo

Memo

Memo